Mini project report on

Digital Museum Artifact & Restoration Management System

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

UE24CS351A – DBMS Project

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Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

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CERTIFICATE

This is to certify that the mini project entitled

Digital Museum Artifact & Restoration Management System

is a bonafide work carried out by

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In partial fulfilment for the completion of fifth semester DBMS Project (UE24CS351A) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2026 – DEC. 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature
Dr. Shilpa S
Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **Digital Museum Artifact & Restoration Management System** has been carried out by us under the guidance of **Dr. Shilpa S, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester AUG – DEC 2026.

Riona Jeena Dsouza	**PES2UG24CS654**	**<Signature>**
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ABSTRACT

Museums hold thousands of artifacts whose records are spread over registers, spreadsheets and photographs. Details such as where an artifact came from, which exhibition it is displayed in, what condition it is in, and what restoration work has been done on it are hard to keep consistent and hard to query when they are kept this way.

This project presents the Digital Museum Artifact & Restoration Management System, a database-driven application that stores and manages this information in one place. The relational database, designed in MySQL, has 17 tables covering museums, collections, creators, artifacts, exhibitions and displays, digital media, artifact history, condition assessments, restoration projects, experts and materials, staff, visitors and visit records. Integrity is enforced with primary keys, foreign keys with explicit referential actions, unique and not-null constraints, and two triggers; two stored functions and two stored procedures encapsulate frequently used calculations and lookups.

A web application sits on top of the database. The backend is a Flask (Python) REST API and the frontend is a single-page application written in HTML, CSS and JavaScript. It provides staff login, a dashboard, an artifact explorer, add / edit / delete / search / sort / paginate screens for all 17 tables, 18 ready-made SQL reports and a page that demonstrates the triggers, functions and procedures. The report describes the requirements, the ER and relational design, the SQL implementation (DDL, DML, queries, triggers, functions and procedures), the application architecture, and the testing carried out.

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1. INTRODUCTION

1.1 Overview

A museum artifact has information beyond its name. It may have an accession number, creator, origin, acquisition history, condition assessments, photographs, exhibition placements and restoration projects. These details are related and need to remain consistent over time.

The Digital Museum Artifact & Restoration Management System provides a unified application for managing this information. MySQL stores structured entities and relationships that require referential integrity. MongoDB stores flexible research, observation, reference, feedback and digital-documentation records whose fields can vary by document.

The project focuses on database design and implementation as well as a web interface through which authorised staff can work with the data.

1.2 Problem Statement

Museum information is distributed across multiple operational areas. Separate manual records can cause duplicate data, inconsistent references and delays when staff need to retrieve information that spans artifacts, exhibitions and restoration work. A common database and application interface are needed to maintain connected records and support routine operations.

1.3 Objectives

- Design an ER model and relational schema that captures museums, collections, artifacts, creators, exhibitions, restoration work, staff and visitors.
- Implement the schema in MySQL with appropriate keys, constraints and referential actions, and load realistic sample data.
- Use SQL queries (WHERE, ORDER BY, GROUP BY, HAVING, aggregates, joins and nested queries) to produce useful reports.
- Use triggers, functions and stored procedures to enforce business rules and reuse common logic inside the database.
- Build a Flask + JavaScript application with staff login so that the database can be used without writing SQL.
- Provide a browser-based application with staff authentication.
- Integrate MongoDB for flexible document collections.

1.4 Scope

The system is intended for museum staff such as curators, archivists, conservators, restoration managers and the museum director. It covers the following areas:

- Catalogue: museums, collections, creators and artifacts, including accession numbers, estimated value and status.
- Exhibitions: planning exhibitions and recording which artifact is shown in which gallery and display case, and for how long.
- Preservation: condition assessments, artifact history, digital media records and restoration projects with experts and materials.
- Visitors: visitor registration and visit records for each museum.

1.5 Technologies Used

Frontend: Next.js 15 with React 19 and TypeScript. Styling and page presentation are implemented in the Next.js application, with the main authenticated workspace in app/page.tsx and shared styles in app/globals.css.

Backend: Next.js API Route Handlers. The API endpoints handle authentication, user management and application operations. They validate requests, call the database layer and return JSON responses. The project does not use Flask, Python, PyMySQL, or a separate Flask REST server.

Relational database: MySQL. The connection is implemented in lib/db.ts using mysql2. The configured database name is supplied through MYSQL_DATABASE in .env.local; for the existing installation used during project setup, this is digital_museum_dbs.

Document database: MongoDB. The MongoDB Node.js driver is used by the application to connect using MONGODB_URI and MONGODB_DATABASE. MongoDB Studio supports five named collections and saves valid JSON documents with createdAt, updatedAt and createdBy metadata.

Authentication: bcryptjs is used for password hashing, and the application uses HTTP-only session cookies. Session and setup secrets are configured through environment variables.

The browser communicates with Next.js API routes. The API layer communicates with MySQL and MongoDB. MySQL remains the source for relational museum records; MongoDB is used for flexible document-oriented records.

2. PROBLEM DEFINITION

Museum information is distributed across several operational areas. Artifact cataloguing, exhibition scheduling, conservation work, staff responsibilities, and visitor records have different data requirements but are connected. Without a common database, duplicate records, inconsistent references, and difficulty retrieving combined information can arise.

Problem statement

Design and implement a relational database and web application that stores museum and artifact information, maintains relationships among collections and operational records, supports restoration and condition tracking, and enables staff to retrieve and update information through SQL queries and a user interface.

2.1 Functional requirements

- Store museum details and their collections.
- Store creator details and associate creators with artifacts when known.
- Maintain artifact identification, accession number, type, origin, dates, value, and condition status.
- Maintain staff credentials and staff details.
- Create exhibitions and associate artifacts with exhibitions through display records.
- Store media files/URLs and artifact history records.
- Record condition assessments and recommended conservation actions.
- Manage restoration projects, managers, experts, materials, and costs.
- Register visitors and record their visits to museums.
- Provide dashboard summaries, filters, reports, and database-feature demonstrations.

2.2 Non-functional considerations

The schema uses primary and foreign keys to preserve referential integrity, unique constraints to avoid duplicate accession numbers and staff identifiers, and transaction rollback for trigger demonstrations. Passwords are stored as salted hashes by the application setup and staff passwords are not returned by the table API. The current project is a local demonstration application; deployment hardening, role-based authorization, backups, and production security configuration would be needed before real-world use.

2.1 Software Requirements

The final package requires Node.js and npm, a running MySQL server, a running MongoDB server, and a modern browser. Visual Studio Code and MySQL Workbench/MongoDB Compass may be used for development and database inspection. The project package declares Next.js, React, TypeScript, mysql2, mongodb and bcryptjs dependencies in package.json.

3. ER MODEL

3.1 Entity–Relationship Model

The main entities are MUSEUM, COLLECTION, CREATOR, ARTIFACT, EXHIBITION, STAFF, RESTORATION_PROJECT, RESTORATION_EXPERT, RESTORATION_MATERIAL, CONDITION_ASSESSMENT, ARTIFACT_HISTORY, ARTIFACT_MEDIA, VISITOR and VISIT_RECORD. DISPLAY is the associative entity between EXHIBITION and ARTIFACT, while PROJECT_EXPERT and PROJECT_MATERIAL resolve the many-to-many relationships of a restoration project with experts and materials.

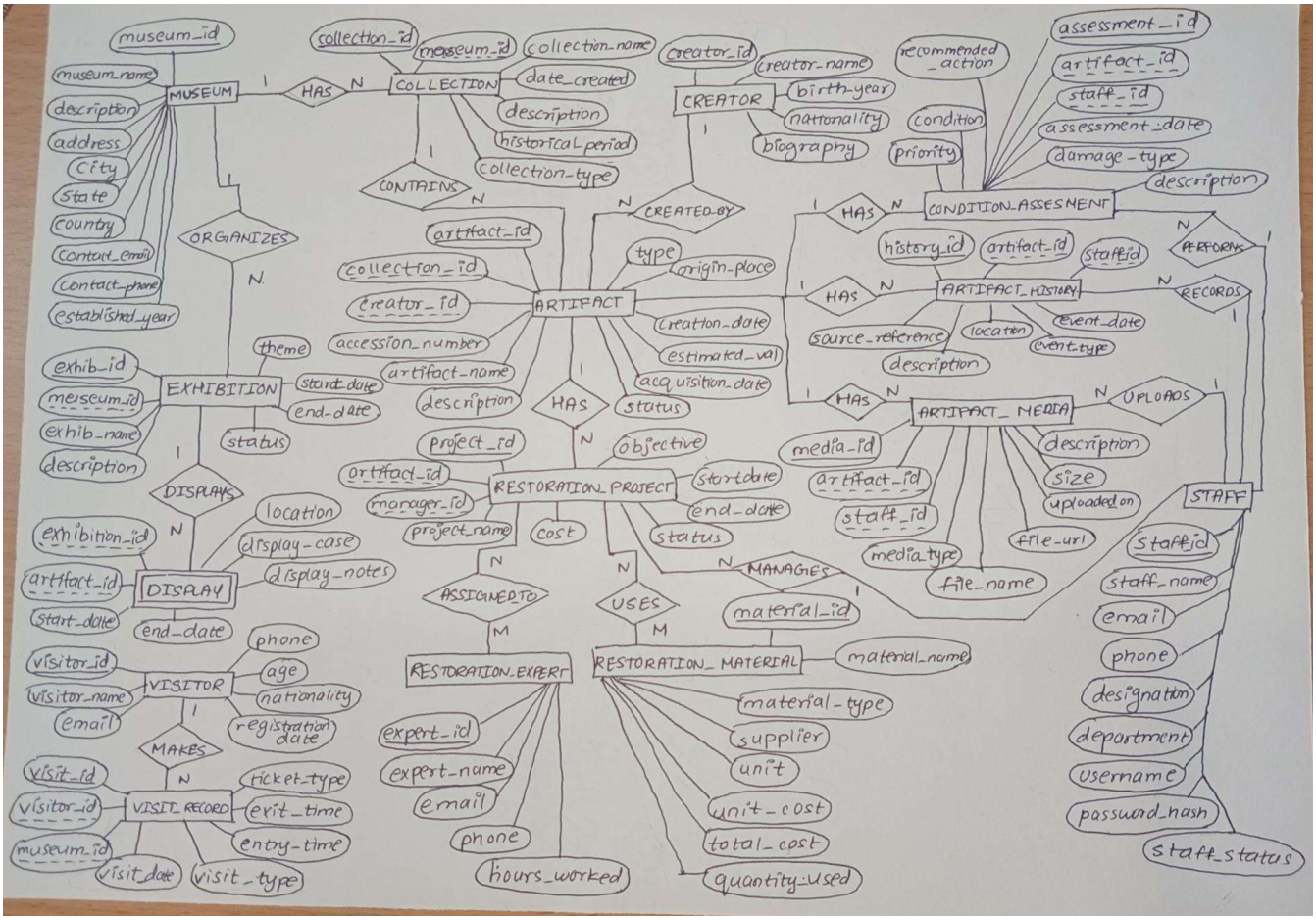


Figure 1: ER diagram of the Digital Museum database

Main entities:

Entity	Purpose
MUSEUM	Stores museum identity, address, contact details, and establishment year.
COLLECTION	Stores collections belonging to a museum.
CREATOR	Stores creator/artist details; creator may be unknown for an artifact.
ARTIFACT	Central catalogue record, linked to a collection and optionally to a creator.
STAFF	Stores staff contact, designation, department, username, and password hash.
EXHIBITION / DISPLAY	Stores exhibition details and the artifacts displayed in each exhibition.
ARTIFACT_MEDIA / ARTIFACT_HISTORY	Stores digital media metadata and historical events for artifacts.
CONDITION_ASSESSMENT	Stores condition, damage, recommended action, and priority.
RESTORATION_PROJECT	Stores restoration objectives, dates, status, cost, and project manager.
RESTORATION_EXPERT / PROJECT_EXPERT	Stores experts and their many-to-many assignment to projects.
RESTORATION_MATERIAL / PROJECT_MATERIAL	Stores restoration materials and their many-to-many association with projects.
VISITOR / VISIT_RECORD	Stores visitor details and visits to museums.

3.2 Relationships

Table 3: Relationships and cardinalities

Relationship	Entities	Cardinality
HAS	MUSEUM – COLLECTION	1 : N
ORGANIZES	MUSEUM – EXHIBITION	1 : N
CONTAINS	COLLECTION – ARTIFACT	1 : N
CREATED_BY	CREATOR – ARTIFACT	1 : N
DISPLAYS	EXHIBITION – ARTIFACT (table DISPLAY)	M : N
HAS	ARTIFACT – RESTORATION_PROJECT	1 : N
HAS	ARTIFACT – CONDITION_ASSESSMENT	1 : N

HAS	ARTIFACT – ARTIFACT_HISTORY	1 : N
HAS	ARTIFACT – ARTIFACT_MEDIA	1 : N
MANAGES	STAFF – RESTORATION_PROJECT	1 : N
PERFORMS	STAFF – CONDITION_ASSESSMENT	1 : N
RECORDS	STAFF – ARTIFACT_HISTORY	1 : N
UPLOADS	STAFF – ARTIFACT_MEDIA	1 : N
ASSIGNED_TO	RESTORATION_PROJECT – RESTORATION_EXPERT (table PROJECT_EXPERT)	M : N
USES	RESTORATION_PROJECT – RESTORATION_MATERIAL (table PROJECT_MATERIAL)	M : N
MAKES	VISITOR – VISIT_RECORD	1 : N
RECEIVES	MUSEUM – VISIT_RECORD	1 : N

4. ER TO RELATIONAL MAPPING

4.1 STEPS OF ALGORITHM FOR CHOOSEN PROBLEM

Mapping the ER diagram gives the relational schema in Figure 2. Primary keys are underlined and arrows show foreign keys.

Table 4: Tables, keys and purpose

Table	**Primary key**	**Foreign keys**	**Purpose**
MUSEUM	museum_id	–	Museum details and contact information
COLLECTION	collection_id	museum_id	Named collections inside a museum
CREATOR	creator_id	–	Artists / artisans (may be "Unknown …")
ARTIFACT	artifact_id	collection_id, creator_id	The artifacts; unique accession_number
STAFF	staff_id	–	Employees and login credentials
EXHIBITION	exhibition_id	museum_id	Exhibitions with theme, dates and status
DISPLAY	(exhibition_id, artifact_id)	exhibition_id, artifact_id	Artifact placed in an exhibition (gallery, case, period)
ARTIFACT_MEDIA	media_id	artifact_id, staff_id	Image / document records of an artifact
ARTIFACT_HISTORY	history_id	artifact_id, staff_id	Dated events such as acquisition or discovery
CONDITION_ASSESSMENT	assessment_id	artifact_id, staff_id	Condition, damage type, priority, recommended action
RESTORATION_PROJECT	project_id	artifact_id, manager_id	Restoration work with dates, status and cost

RESTORATION_EXPERT	expert_id	–	Conservators / specialists and hours worked
PROJECT_EXPERT	(project_id, expert_id)	project_id, expert_id	Experts assigned to a project
RESTORATION_MATERIAL	material_id	–	Materials with unit cost, quantity and total cost
PROJECT_MATERIAL	(project_id, material_id)	project_id, material_id	Materials used in a project
VISITOR	visitor_id	–	Registered visitors
VISIT_RECORD	visit_id	visitor_id, museum_id	One visit to one museum

4.1.1 Integrity Constraints and Referential Actions

- **Primary keys:** every table has one; the junction tables DISPLAY, PROJECT_EXPERT and PROJECT_MATERIAL use composite keys so the same pair cannot be stored twice.
- **NOT NULL:** names, accession numbers, assessment and event dates, visit dates, and staff username, e-mail and password hash.
- **UNIQUE:** ARTIFACT.accession_number, STAFF.email and STAFF.username.
- **ON DELETE CASCADE:** child rows that make no sense without their parent are removed with it (collections of a museum, exhibitions of a museum, display / media / history / assessment / restoration rows of an artifact, junction rows, and visit records).
- **ON DELETE RESTRICT:** ARTIFACT.collection_id – a collection that still contains artifacts cannot be deleted, protecting the catalogue from accidental loss.
- **ON DELETE SET NULL:** optional references such as ARTIFACT.creator_id and the staff_id / manager_id columns – deleting a staff member or creator keeps the artifact and its records.
- **ON UPDATE CASCADE:** used on every foreign key so that a changed key value propagates to child tables.

4.1.2 Normalisation

Every table has atomic columns and a primary key (1NF). In the junction tables the non-key attributes depend on the whole composite key (2NF); the other tables have single-column keys, so partial dependencies cannot occur. Descriptive attributes depend only on the key of their own table: museum details are held once in MUSEUM and referred to by museum_id, creator details once in CREATOR, and staff details once in STAFF, which removes transitive dependencies (3NF). Many-to-many relationships are resolved with separate tables instead of repeating groups.

One attribute is derived on purpose: RESTORATION_MATERIAL.total_cost equals unit_cost × quantity_used. It is kept so that cost reports are simple, and a trigger (Section 4.4) guarantees that it can never disagree with its inputs.

4.1.3 Changes Made While Building the Application

While connecting the application to the database from Review 2 a few small adjustments were made to the SQL. They are all included in the sql/ folder of the project.

- CREATOR.birth_year was changed from YEAR to INT, because the MySQL YEAR type cannot store years such as 1100 or 1200 that occur in the data.
- CREATOR, EXHIBITION and ARTIFACT_MEDIA identifiers were made AUTO_INCREMENT so that the application can add rows without supplying an id.
- DISPLAY, PROJECT_EXPERT and PROJECT_MATERIAL received composite primary keys, so duplicate rows are impossible and rows can be edited or deleted from the interface.
- The two functions use READS SQL DATA instead of DETERMINISTIC (they read tables and depend on the current date).
- BEFORE UPDATE versions of the material-cost and exhibition-date triggers were added so that edits made from the application are validated as well.

4.2 COMPLETE DIAGRAM OF RELATIONAL MAPPING

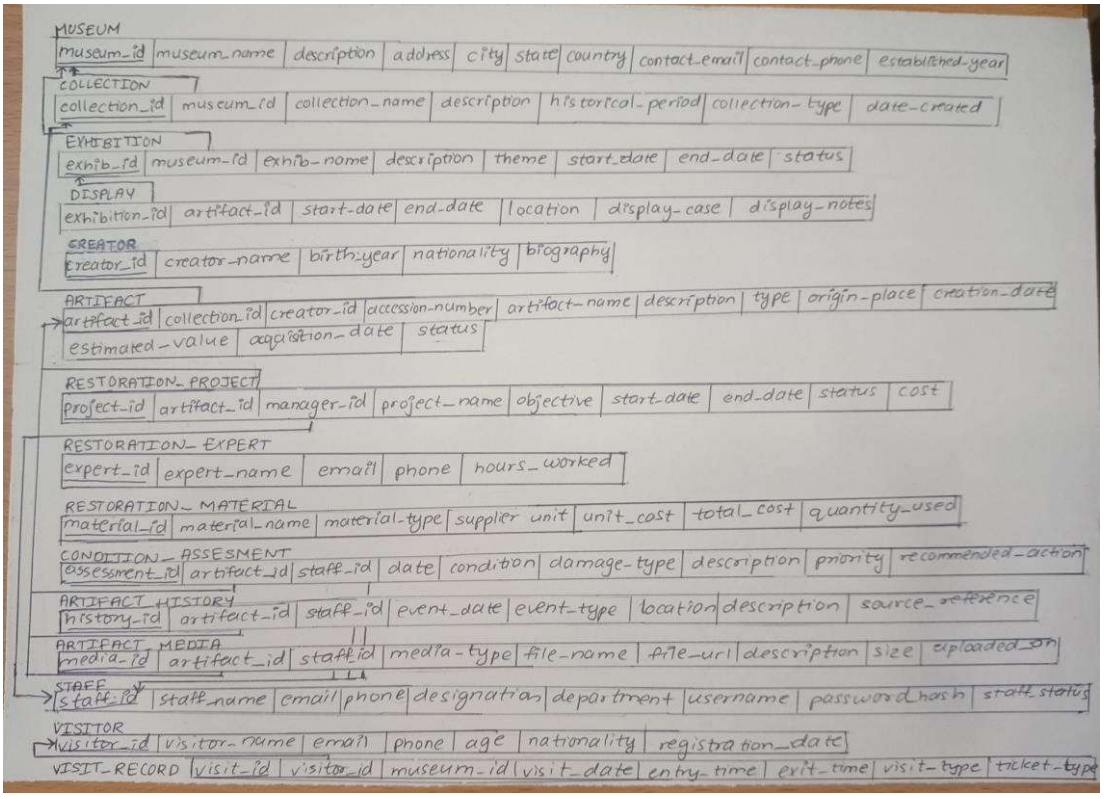


Figure 2: Relational schema diagram

5. DDL STATEMENTS

DDL (Data Definition Language) statements define the database structure. The implementation creates the database digital_museum_db and defines 17 tables. The following extracts show the table-creation pattern; the complete schema is present in digital_museum/sql/schema.sql in the submitted application archive.

Database creation:

```
mysql> CREATE DATABASE digital_museum_db;
Query OK, 1 row affected (0.05 sec)

mysql> USE digital_museum_db;
Database changed
```

Table creation:

```
mysql> CREATE TABLE MUSEUM (
    ->     museum_id INT AUTO_INCREMENT,
    ->     museum_name VARCHAR(150) NOT NULL,
    ->     description TEXT,
    ->     address VARCHAR(255),
    ->     city VARCHAR(100),
    ->     state VARCHAR(100),
    ->     country VARCHAR(100),
    ->     contact_email VARCHAR(150),
    ->     contact_phone VARCHAR(30),
    ->     established_year YEAR,
    ->     PRIMARY KEY (museum_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE COLLECTION (
    ->     collection_id INT AUTO_INCREMENT,
    ->     museum_id INT NOT NULL,
    ->     collection_name VARCHAR(150) NOT NULL,
    ->     description TEXT,
    ->     historical_period VARCHAR(100),
    ->     collection_type VARCHAR(100),
    ->     date_created DATE,
    ->     PRIMARY KEY (collection_id),
    ->     CONSTRAINT fk_collection_museum FOREIGN KEY (museum_id) REFERENCE
S MUSEUM(museum_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE CREATOR (
    ->     creator_id INT AUTO_INCREMENT,
    ->     creator_name VARCHAR(150) NOT NULL,
    ->     birth_year INT,
    ->     nationality VARCHAR(100),
    ->     biography TEXT,
    ->     PRIMARY KEY (creator_id)
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE ARTIFACT (
    ->     artifact_id INT AUTO_INCREMENT,
    ->     collection_id INT NOT NULL,
    ->     creator_id INT,
    ->     accession_number VARCHAR(100) NOT NULL,
    ->     artifact_name VARCHAR(200) NOT NULL,
    ->     description TEXT,
    ->     artifact_type VARCHAR(100),
    ->     origin_place VARCHAR(150),
    ->     creation_date DATE,
    ->     estimated_value DECIMAL(15,2),
    ->     acquisition_date DATE,
    ->     status VARCHAR(100),
    ->     PRIMARY KEY (artifact_id),
    ->     CONSTRAINT uq_artifact_accession UNIQUE (accession_number),
    ->     CONSTRAINT fk_artifact_collection FOREIGN KEY (collection_id) REF
ERENCES COLLECTION(collection_id)
    ->         ON UPDATE CASCADE ON DELETE RESTRICT,
    ->     CONSTRAINT fk_artifact_creator FOREIGN KEY (creator_id) REFERENCE
S CREATOR(creator_id)
    ->         ON UPDATE CASCADE ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE STAFF (
    ->     staff_id INT AUTO_INCREMENT,
    ->     staff_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150) NOT NULL,
    ->     phone VARCHAR(30),
    ->     designation VARCHAR(100),
    ->     department VARCHAR(100),
    ->     username VARCHAR(100) NOT NULL,
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     staff_status VARCHAR(50) DEFAULT 'Active',
    ->     role ENUM('ADMIN','STAFF') NOT NULL DEFAULT 'STAFF',
    ->     PRIMARY KEY (staff_id),
    ->     CONSTRAINT uq_staff_email UNIQUE (email),
    ->     CONSTRAINT uq_staff_username UNIQUE (username)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE EXHIBITION (
    ->     exhibition_id INT AUTO_INCREMENT,
    ->     museum_id INT NOT NULL,
    ->     exhibition_name VARCHAR(200) NOT NULL,
    ->     description TEXT,
    ->     theme VARCHAR(150),
    ->     start_date DATE,
    ->     end_date DATE,
    ->     status VARCHAR(50),
    ->     PRIMARY KEY (exhibition_id),
    ->     CONSTRAINT fk_exhibition_museum FOREIGN KEY (museum_id) REFERENCE
S MUSEUM(museum_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE DISPLAY (
    ->     exhibition_id INT NOT NULL,
    ->     artifact_id INT NOT NULL,
    ->     display_start_date DATE,
    ->     display_end_date DATE,
    ->     display_location VARCHAR(150),
    ->     display_case VARCHAR(100),
    ->     display_notes TEXT,
    ->     PRIMARY KEY (exhibition_id, artifact_id),
    ->     CONSTRAINT fk_display_exhibition FOREIGN KEY (exhibition_id) REFE
RENCES EXHIBITION(exhibition_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_display_artifact FOREIGN KEY (artifact_id) REFERENC
ES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE ARTIFACT_MEDIA (
    ->     media_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     media_type VARCHAR(50),
    ->     file_name VARCHAR(255),
    ->     file_url VARCHAR(500),
    ->     description TEXT,
    ->     uploaded_on DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     file_size BIGINT,
    ->     PRIMARY KEY (media_id),
    ->     CONSTRAINT fk_media_artifact FOREIGN KEY (artifact_id) REFERENCES
 ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_media_staff FOREIGN KEY (staff_id) REFERENCES STAFF
(staff_id)
    ->         ON UPDATE CASCADE ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE ARTIFACT_HISTORY (
    ->     history_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     event_date DATE NOT NULL,
    ->     event_type VARCHAR(100),
    ->     location VARCHAR(150),
    ->     description TEXT,
    ->     source_reference VARCHAR(500),
    ->     PRIMARY KEY (history_id),
    ->     CONSTRAINT fk_history_artifact FOREIGN KEY (artifact_id) REFERENC
ES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_history_staff FOREIGN KEY (staff_id) REFERENCES STA
FF(staff_id)
    ->         ON UPDATE CASCADE ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE RESTORATION_PROJECT (
    ->     project_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     manager_id INT,
    ->     project_name VARCHAR(200) NOT NULL,
    ->     objective TEXT,
    ->     start_date DATE,
    ->     end_date DATE,
    ->     status VARCHAR(50),
    ->     cost DECIMAL(15,2),
    ->     PRIMARY KEY (project_id),
    ->     CONSTRAINT fk_project_artifact FOREIGN KEY (artifact_id) REFERENC
ES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_project_manager FOREIGN KEY (manager_id) REFERENCES
 STAFF(staff_id)
    ->         ON UPDATE CASCADE ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE CONDITION_ASSESSMENT (
    ->     assessment_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     assessment_date DATE NOT NULL,
    ->     conditions VARCHAR(100),
    ->     damage_type VARCHAR(100),
    ->     damage_description TEXT,
    ->     recommended_action TEXT,
    ->     priority VARCHAR(50),
    ->     PRIMARY KEY (assessment_id),
    ->     CONSTRAINT fk_assessment_artifact FOREIGN KEY (artifact_id) REFER
ENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_assessment_staff FOREIGN KEY (staff_id) REFERENCES
 STAFF(staff_id)
    ->         ON UPDATE CASCADE ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE RESTORATION_EXPERT (
    ->     expert_id INT AUTO_INCREMENT,
    ->     expert_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150),
    ->     phone VARCHAR(30),
    ->     hours_worked DECIMAL(8,2),
    ->     PRIMARY KEY (expert_id)
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE PROJECT_EXPERT (
    ->     project_id INT NOT NULL,
    ->     expert_id INT NOT NULL,
    ->     PRIMARY KEY (project_id, expert_id),
    ->     CONSTRAINT fk_project_expert_project FOREIGN KEY (project_id) REF
ERENCES RESTORATION_PROJECT(project_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_project_expert_expert FOREIGN KEY (expert_id) REFER
ENCES RESTORATION_EXPERT(expert_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE RESTORATION_MATERIAL (
    ->     material_id INT AUTO_INCREMENT,
    ->     material_name VARCHAR(150) NOT NULL,
    ->     material_type VARCHAR(100),
    ->     supplier VARCHAR(150),
    ->     unit VARCHAR(50),
    ->     unit_cost DECIMAL(12,2),
    ->     quantity_used DECIMAL(10,2),
    ->     total_cost DECIMAL(15,2),
    ->     PRIMARY KEY (material_id)
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE PROJECT_MATERIAL (
    ->     project_id INT NOT NULL,
    ->     material_id INT NOT NULL,
    ->     PRIMARY KEY (project_id, material_id),
    ->     CONSTRAINT fk_project_material_project FOREIGN KEY (project_id) R
EFERENCES RESTORATION_PROJECT(project_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_project_material_material FOREIGN KEY (material_id)
 REFERENCES RESTORATION_MATERIAL(material_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE VISITOR (
    ->     visitor_id INT AUTO_INCREMENT,
    ->     visitor_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150),
    ->     phone VARCHAR(30),
    ->     age INT,
    ->     nationality VARCHAR(100),
    ->     registration_date DATE,
    ->     PRIMARY KEY (visitor_id)
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE VISIT_RECORD (
    ->     visit_id INT AUTO_INCREMENT,
    ->     visitor_id INT NOT NULL,
    ->     museum_id INT NOT NULL,
    ->     visit_date DATE NOT NULL,
    ->     entry_time TIME,
    ->     exit_time TIME,
    ->     visit_type VARCHAR(50),
    ->     ticket_type VARCHAR(50),
    ->     PRIMARY KEY (visit_id),
    ->     CONSTRAINT fk_visit_visitor FOREIGN KEY (visitor_id) REFERENCES V
ISITOR(visitor_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE,
    ->     CONSTRAINT fk_visit_museum FOREIGN KEY (museum_id) REFERENCES MUS
EUM(museum_id)
    ->         ON UPDATE CASCADE ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE INDEX idx_artifact_name ON ARTIFACT(artifact_name);
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_artifact_status_type ON ARTIFACT(status, artifact_ty
pe);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_exhibition_dates ON EXHIBITION(start_date, end_date)
;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_visit_museum_date ON VISIT_RECORD(museum_id, visit_d
ate);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_history_artifact_date ON ARTIFACT_HISTORY(artifact_i
d, event_date);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> show tables;
+----------------------+
| Tables_in_ex         |
+----------------------+
| artifact             |
| artifact_history     |
| artifact_media       |
| collection           |
| condition_assessment |
| creator              |
| display              |
| exhibition           |
| museum               |
| project_expert       |
| project_material     |
| restoration_expert   |
| restoration_material |
| restoration_project  |
| staff                |
| visit_record         |
| visitor              |
+----------------------+
17 rows in set (0.01 sec)
```

```
mysql> desc artifact;
+------------------+---------------+------+-----+---------+----------------+
| Field            | Type          | Null | Key | Default | Extra          |
+------------------+---------------+------+-----+---------+----------------+
| artifact_id      | int           | NO   | PRI | NULL    | auto_increment |
| collection_id    | int           | NO   | MUL | NULL    |                |
| creator_id       | int           | YES  | MUL | NULL    |                |
| accession_number | varchar(100)  | NO   | UNI | NULL    |                |
| artifact_name    | varchar(200)  | NO   | MUL | NULL    |                |
| description      | text          | YES  |     | NULL    |                |
| artifact_type    | varchar(100)  | YES  |     | NULL    |                |
| origin_place     | varchar(150)  | YES  |     | NULL    |                |
| creation_date    | date          | YES  |     | NULL    |                |
| estimated_value  | decimal(15,2) | YES  |     | NULL    |                |
| acquisition_date | date          | YES  |     | NULL    |                |
| status           | varchar(100)  | YES  | MUL | NULL    |                |
+------------------+---------------+------+-----+---------+----------------+
12 rows in set (0.00 sec)

mysql> desc artifact_history;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| history_id       | int          | NO   | PRI | NULL    | auto_increment |
| artifact_id      | int          | NO   | MUL | NULL    |                |
| staff_id         | int          | YES  | MUL | NULL    |                |
| event_date       | date         | NO   |     | NULL    |                |
| event_type       | varchar(100) | YES  |     | NULL    |                |
| location         | varchar(150) | YES  |     | NULL    |                |
| description      | text         | YES  |     | NULL    |                |
| source_reference | varchar(500) | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
mysql> desc artifact_media;
+-------------+--------------+------+-----+-------------------+-------------------+
| Field       | Type         | Null | Key | Default           | Extra             |
+-------------+--------------+------+-----+-------------------+-------------------+
| media_id    | int          | NO   | PRI | NULL              | auto_increment    |
| artifact_id | int          | NO   | MUL | NULL              |                   |
| staff_id    | int          | YES  | MUL | NULL              |                   |
| media_type  | varchar(50)  | YES  |     | NULL              |                   |
| file_name   | varchar(255) | YES  |     | NULL              |                   |
| file_url    | varchar(500) | YES  |     | NULL              |                   |
| description | text         | YES  |     | NULL              |                   |
| uploaded_on | datetime     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| file_size   | bigint       | YES  |     | NULL              |                   |
+-------------+--------------+------+-----+-------------------+-------------------+
9 rows in set (0.00 sec)

mysql> desc collection;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| collection_id     | int          | NO   | PRI | NULL    | auto_increment |
| museum_id         | int          | NO   | MUL | NULL    |                |
| collection_name   | varchar(150) | NO   |     | NULL    |                |
| description       | text         | YES  |     | NULL    |                |
| historical_period | varchar(100) | YES  |     | NULL    |                |
| collection_type   | varchar(100) | YES  |     | NULL    |                |
| date_created      | date         | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

```
mysql> desc condition_assessment;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| assessment_id      | int          | NO   | PRI | NULL    | auto_increment |
| artifact_id        | int          | NO   | MUL | NULL    |                |
| staff_id           | int          | YES  | MUL | NULL    |                |
| assessment_date    | date         | NO   |     | NULL    |                |
| conditions         | varchar(100) | YES  |     | NULL    |                |
| damage_type        | varchar(100) | YES  |     | NULL    |                |
| damage_description | text         | YES  |     | NULL    |                |
| recommended_action | text         | YES  |     | NULL    |                |
| priority           | varchar(50)  | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)

mysql> desc creator;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| creator_id   | int          | NO   | PRI | NULL    | auto_increment |
| creator_name | varchar(150) | NO   |     | NULL    |                |
| birth_year   | int          | YES  |     | NULL    |                |
| nationality  | varchar(100) | YES  |     | NULL    |                |
| biography    | text         | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql> desc display;
+--------------------+--------------+------+-----+---------+-------+
| Field              | Type         | Null | Key | Default | Extra |
+--------------------+--------------+------+-----+---------+-------+
| exhibition_id      | int          | NO   | PRI | NULL    |       |
| artifact_id        | int          | NO   | PRI | NULL    |       |
| display_start_date | date         | YES  |     | NULL    |       |
| display_end_date   | date         | YES  |     | NULL    |       |
| display_location   | varchar(150) | YES  |     | NULL    |       |
| display_case       | varchar(100) | YES  |     | NULL    |       |
| display_notes      | text         | YES  |     | NULL    |       |
+--------------------+--------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

```
mysql> desc exhibition;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| exhibition_id   | int          | NO   | PRI | NULL    | auto_increment |
| museum_id       | int          | NO   | MUL | NULL    |                |
| exhibition_name | varchar(200) | NO   |     | NULL    |                |
| description     | text         | YES  |     | NULL    |                |
| theme           | varchar(150) | YES  |     | NULL    |                |
| start_date      | date         | YES  | MUL | NULL    |                |
| end_date        | date         | YES  |     | NULL    |                |
| status          | varchar(50)  | YES  |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)

mysql> desc museum;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| museum_id        | int          | NO   | PRI | NULL    | auto_increment |
| museum_name      | varchar(150) | NO   |     | NULL    |                |
| description      | text         | YES  |     | NULL    |                |
| address          | varchar(255) | YES  |     | NULL    |                |
| city             | varchar(100) | YES  |     | NULL    |                |
| state            | varchar(100) | YES  |     | NULL    |                |
| country          | varchar(100) | YES  |     | NULL    |                |
| contact_email    | varchar(150) | YES  |     | NULL    |                |
| contact_phone    | varchar(30)  | YES  |     | NULL    |                |
| established_year | year         | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
10 rows in set (0.00 sec)

mysql> desc project_expert;
+------------+------+------+-----+---------+-------+
| Field      | Type | Null | Key | Default | Extra |
+------------+------+------+-----+---------+-------+
| project_id | int  | NO   | PRI | NULL    |       |
| expert_id  | int  | NO   | PRI | NULL    |       |
+------------+------+------+-----+---------+-------+
```

```
mysql> desc project_material;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| project_id  | int  | NO   | PRI | NULL    |       |
| material_id | int  | NO   | PRI | NULL    |       |
+-------------+------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> desc restoration_expert;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| expert_id    | int          | NO   | PRI | NULL    | auto_increment |
| expert_name  | varchar(150) | NO   |     | NULL    |                |
| email        | varchar(150) | YES  |     | NULL    |                |
| phone        | varchar(30)  | YES  |     | NULL    |                |
| hours_worked | decimal(8,2) | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql> desc restoration_project;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| project_id   | int           | NO   | PRI | NULL    | auto_increment |
| artifact_id  | int           | NO   | MUL | NULL    |                |
| manager_id   | int           | YES  | MUL | NULL    |                |
| project_name | varchar(200)  | NO   |     | NULL    |                |
| objective    | text          | YES  |     | NULL    |                |
| start_date   | date          | YES  |     | NULL    |                |
| end_date     | date          | YES  |     | NULL    |                |
| status       | varchar(50)   | YES  |     | NULL    |                |
| cost         | decimal(15,2) | YES  |     | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

```
mysql> desc restoration_material;
+---------------+---------------+------+-----+---------+----------------+
| Field         | Type          | Null | Key | Default | Extra          |
+---------------+---------------+------+-----+---------+----------------+
| material_id   | int           | NO   | PRI | NULL    | auto_increment |
| material_name | varchar(150)  | NO   |     | NULL    |                |
| material_type | varchar(100)  | YES  |     | NULL    |                |
| supplier      | varchar(150)  | YES  |     | NULL    |                |
| unit          | varchar(50)   | YES  |     | NULL    |                |
| unit_cost     | decimal(12,2) | YES  |     | NULL    |                |
| quantity_used | decimal(10,2) | YES  |     | NULL    |                |
| total_cost    | decimal(15,2) | YES  |     | NULL    |                |
+---------------+---------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)

mysql> desc staff;
+---------------+----------------------+------+-----+---------+----------------+
| Field         | Type                 | Null | Key | Default | Extra          |
+---------------+----------------------+------+-----+---------+----------------+
| staff_id      | int                  | NO   | PRI | NULL    | auto_increment |
| staff_name    | varchar(150)         | NO   |     | NULL    |                |
| email         | varchar(150)         | NO   | UNI | NULL    |                |
| phone         | varchar(30)          | YES  |     | NULL    |                |
| designation   | varchar(100)         | YES  |     | NULL    |                |
| department    | varchar(100)         | YES  |     | NULL    |                |
| username      | varchar(100)         | NO   | UNI | NULL    |                |
| password_hash | varchar(255)         | NO   |     | NULL    |                |
| staff_status  | varchar(50)          | YES  |     | Active  |                |
| role          | enum('ADMIN','STAFF') | NO   |     | STAFF   |                |
+---------------+----------------------+------+-----+---------+----------------+
10 rows in set (0.00 sec)
```

```
mysql> desc visit_record;
+-------------+-------------+------+-----+---------+----------------+
| Field       | Type        | Null | Key | Default | Extra          |
+-------------+-------------+------+-----+---------+----------------+
| visit_id    | int         | NO   | PRI | NULL    | auto_increment |
| visitor_id  | int         | NO   | MUL | NULL    |                |
| museum_id   | int         | NO   | MUL | NULL    |                |
| visit_date  | date        | NO   |     | NULL    |                |
| entry_time  | time        | YES  |     | NULL    |                |
| exit_time   | time        | YES  |     | NULL    |                |
| visit_type  | varchar(50) | YES  |     | NULL    |                |
| ticket_type | varchar(50) | YES  |     | NULL    |                |
+-------------+-------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)

mysql> desc visitor;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| visitor_id       | int          | NO   | PRI | NULL    | auto_increment |
| visitor_name     | varchar(150) | NO   |     | NULL    |                |
| email            | varchar(150) | YES  |     | NULL    |                |
| phone            | varchar(30)  | YES  |     | NULL    |                |
| age              | int          | YES  |     | NULL    |                |
| nationality      | varchar(100) | YES  |     | NULL    |                |
| registration_date | date        | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

Database initialization

The init_db.py script connects to MySQL, drops and recreates the configured database, executes schema.sql and data.sql, replaces demonstration password placeholders with Werkzeug-generated password hashes, and then creates the triggers, functions, and procedures from routines.sql. Because the script drops the configured database, it should only be used when resetting the demonstration database.

6. DML STATEMENTS

DML (Data Manipulation Language) statements insert, update, and delete records. The data.sql file contains demonstration records for the 17 tables. The SQL script includes sample rows for museums, collections, creators, staff, artifacts, exhibitions, displays, media, artifact history, restoration records, experts, materials, visitors, and visits.

INSERT statements:

```
mysql> INSERT INTO MUSEUM (museum_id, museum_name, description, address, city, state, country, contact_email, contact_phone, established_year) VALUES
    -> (1, 'National Heritage Museum', 'Museum preserving Indian historical and cultural artifacts.', 'MG Road', 'Bengaluru', 'Karnataka', 'India', 'info@nationalheritage.com', '0804000100
1', 1985),
    -> (2, 'Royal History Museum', 'Museum dedicated to royal history and traditional artifacts.', 'Palace Road', 'Mysuru', 'Karnataka', 'India', 'info@royalhistory.com', '08214000102', 19
92),
    -> (3, 'Indian Art Museum', 'Museum showcasing traditional and modern Indian art.', 'Museum Road', 'New Delhi', 'Delhi', 'India', 'info@indianartmuseum.com', '01140001003', 1978),
    -> (4, 'South Indian Heritage Museum', 'Museum preserving cultural heritage from South India.', 'Temple Street', 'Chennai', 'Tamil Nadu', 'India', 'info@sihmuseum.com', '04440001004',
1995),
    -> (5, 'Ancient Civilisation Museum', 'Museum containing archaeological and ancient civilisation artifacts.', 'Heritage Avenue', 'Hyderabad', 'Telangana', 'India', 'info@ancientcivilis
ation.com', '04040001005', 2001);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO COLLECTION (collection_id, museum_id, collection_name, description, historical_period, collection_type, date_created) VALUES
    -> (1, 1, 'Ancient Indian Art', 'Collection of ancient Indian sculptures and artwork.', 'Ancient India', 'Sculpture', '1986-01-10'),
    -> (2, 1, 'Medieval Manuscripts', 'Collection of historical manuscripts and documents.', 'Medieval Period', 'Manuscript', '1987-05-15'),
    -> (3, 2, 'Royal Heritage', 'Artifacts belonging to historical royal families.', 'Medieval India', 'Royal Artifacts', '1993-03-20'),
    -> (4, 2, 'Mysuru Palace Collection', 'Objects associated with Mysuru royal history.', '18th-20th Century', 'Royal Collection', '1994-04-15'),
    -> (5, 3, 'Indian Paintings', 'Collection of traditional Indian paintings.', '19th-20th Century', 'Painting', '1979-02-12'),
    -> (6, 3, 'Modern Art', 'Collection of modern Indian artworks.', '20th-21st Century', 'Modern Art', '1985-07-21'),
    -> (7, 4, 'Temple Artifacts', 'Artifacts collected from South Indian temples.', 'Ancient-Medieval', 'Religious Art', '1996-03-10'),
    -> (8, 4, 'Traditional Crafts', 'Traditional crafts and handmade cultural objects.', '19th-20th Century', 'Craft', '1997-08-18'),
    -> (9, 5, 'Archaeological Collection', 'Artifacts discovered through archaeological research.', 'Ancient Period', 'Archaeological', '2002-01-20'),
    -> (10, 5, 'Coins and Currency', 'Historical coins and currency from different periods.', 'Ancient-Modern', 'Numismatics', '2003-06-11');
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO CREATOR (creator_id, creator_name, birth_year, nationality, biography) VALUES
    -> (1, 'Ravi Varma', 1848, 'Indian', 'Indian painter known for paintings based on Indian mythology.'),
    -> (2, 'Unknown Ancient Sculptor', 1200, 'Indian', 'Ancient sculptor whose identity remains unknown.'),
    -> (3, 'Mysuru Royal Artisan', 1750, 'Indian', 'Traditional artisan associated with the Mysuru royal court.'),
    -> (4, 'Unknown Manuscript Artist', 1600, 'Indian', 'Artist responsible for illustrating historical manuscripts.'),
    -> (5, 'Abanindranath Tagore', 1871, 'Indian', 'Indian painter and one of the founders of the Bengal School of Art.'),
    -> (6, 'Nandalal Bose', 1882, 'Indian', 'Indian painter associated with the Bengal School of Art.'),
    -> (7, 'Jamini Roy', 1887, 'Indian', 'Indian painter known for folk-inspired artistic styles.'),
    -> (8, 'Unknown Temple Artisan', 1400, 'Indian', 'Traditional artisan responsible for temple artwork.'),
    -> (9, 'Ancient Metalworker', 1100, 'Indian', 'Ancient artisan specializing in metal objects.'),
    -> (10, 'Royal Coin Maker', 1650, 'Indian', 'Artisan associated with historical royal coin production.');
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
mysql> -- password_hash values are placeholders ("hash_<password>"); init_db.py replaces them with real hashes.
mysql> INSERT INTO STAFF (staff_id, staff_name, email, phone, designation, department, username, password_hash, staff_status) VALUES
    -> (1, 'Anita Rao', 'anita@museum.com', '9876500001', 'Curator', 'Collections', 'anita.rao', 'hash_anita123', 'Active'),
    -> (2, 'Rahul Sharma', 'rahul@museum.com', '9876500002', 'Restoration Manager', 'Restoration', 'rahul.sharma', 'hash_rahul123', 'Active'),
    -> (3, 'Meera Nair', 'meera@museum.com', '9876500003', 'Archivist', 'Archives', 'meera.nair', 'hash_meera123', 'Active'),
    -> (4, 'Kiran Kumar', 'kiran@museum.com', '9876500004', 'Museum Director', 'Administration', 'kiran.kumar', 'hash_kiran123', 'Active'),
    -> (5, 'Priya Menon', 'priya@museum.com', '9876500005', 'Conservator', 'Restoration', 'priya.menon', 'hash_priya123', 'Active'),
    -> (6, 'Vivek Rao', 'vivek@museum.com', '9876500006', 'Researcher', 'Research', 'vivek.rao', 'hash_vivek123', 'Active'),
    -> (7, 'Sneha Iyer', 'sneha@museum.com', '9876500007', 'Curator', 'Exhibitions', 'sneha.iyer', 'hash_sneha123', 'Active'),
    -> (8, 'Arjun Das', 'arjun@museum.com', '9876500008', 'Security Officer', 'Security', 'arjun.das', 'hash_arjun123', 'Active'),
    -> (9, 'Divya Shah', 'divya@museum.com', '9876500009', 'Archivist', 'Archives', 'divya.shah', 'hash_divya123', 'Active'),
    -> (10, 'Manoj Kumar', 'manoj@museum.com', '9876500010', 'Technician', 'Digital Archives', 'manoj.kumar', 'hash_manoj123', 'Active');
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO ARTIFACT (artifact_id, collection_id, creator_id, accession_number, artifact_name, description, artifact_type, origin_place, creation_date, estimated_value, acquisition_
date, status) VALUES
    -> (1, 1, 2, 'ART-001', 'Ancient Stone Sculpture', 'Stone sculpture depicting an ancient Indian deity.', 'Sculpture', 'Hampi', '1250-01-01', 250000.00, '1986-03-12', 'Good'),
    -> (2, 1, 2, 'ART-002', 'Granite Temple Statue', 'Carved granite statue from a historic temple.', 'Sculpture', 'Hampi', '1300-01-01', 320000.00, '1987-04-15', 'Good'),
    -> (3, 2, 4, 'ART-003', 'Royal Manuscript', 'Handwritten manuscript containing historical records.', 'Manuscript', 'Mysuru', '1650-01-01', 150000.00, '1988-04-10', 'Good'),
    -> (4, 2, 4, 'ART-004', 'Historical Document', 'Historical administrative document.', 'Document', 'Karnataka', '1700-01-01', 80000.00, '1990-08-19', 'Good'),
    -> (5, 3, 3, 'ART-005', 'Royal Wooden Chest', 'Decorated wooden chest used by a historical royal family.', 'Furniture', 'Mysuru', '1780-01-01', 180000.00, '1994-02-15', 'Damaged'),
    -> (6, 3, 3, 'ART-006', 'Royal Sword', 'Ceremonial sword belonging to a royal collection.', 'Weapon', 'Mysuru', '1775-01-01', 300000.00, '1995-07-11', 'Good'),
    -> (7, 4, 3, 'ART-007', 'Royal Throne', 'Decorated wooden throne used for ceremonial purposes.', 'Furniture', 'Mysuru', '1800-01-01', 550000.00, '1996-01-20', 'Needs Restoration'),
    -> (8, 4, 3, 'ART-008', 'Royal Jewellery Box', 'Ornamented jewellery box from the royal collection.', 'Furniture', 'Mysuru', '1820-01-01', 210000.00, '1997-05-12', 'Good'),
    -> (9, 5, 1, 'ART-009', 'Mythological Painting', 'Painting depicting an Indian mythological scene.', 'Painting', 'Mysuru', '1890-01-01', 450000.00, '1987-06-20', 'Needs Restoration'),
    -> (10, 5, 5, 'ART-010', 'Bharat Mata Painting', 'Traditional Indian painting representing Bharat Mata.', 'Painting', 'Kolkata', '1905-01-01', 500000.00, '1990-02-11', 'Good'),
    -> (11, 5, 6, 'ART-011', 'Village Life Painting', 'Painting depicting rural Indian life.', 'Painting', 'Kolkata', '1920-01-01', 350000.00, '1995-03-22', 'Good'),
    -> (12, 6, 7, 'ART-012', 'Folk Art Composition', 'Folk-inspired modern Indian artwork.', 'Painting', 'West Bengal', '1935-01-01', 280000.00, '2000-07-15', 'Good'),
    -> (13, 7, 8, 'ART-013', 'Temple Bronze Idol', 'Traditional bronze idol from a South Indian temple.', 'Religious Artifact', 'Tamil Nadu', '1450-01-01', 400000.00, '1998-04-18', 'Damage
d'),
    -> (14, 7, 8, 'ART-014', 'Temple Stone Panel', 'Decorative stone panel from an ancient temple.', 'Sculpture', 'Tamil Nadu', '1500-01-01', 275000.00, '1999-08-20', 'Good'),
    -> (15, 8, 8, 'ART-015', 'Traditional Wooden Doll', 'Traditional South Indian wooden craft.', 'Craft', 'Karnataka', '1880-01-01', 65000.00, '2001-05-10', 'Good'),
    -> (16, 8, 8, 'ART-016', 'Handwoven Textile', 'Traditional handwoven textile with cultural patterns.', 'Textile', 'Tamil Nadu', '1900-01-01', 95000.00, '2002-09-14', 'Needs Restoration
'),
    -> (17, 9, 9, 'ART-017', 'Ancient Bronze Vessel', 'Bronze vessel discovered during archaeological research.', 'Vessel', 'Andhra Pradesh', '1100-01-01', 600000.00, '2003-01-22', 'Good')
,
    -> (18, 9, 9, 'ART-018', 'Ancient Terracotta Figure', 'Terracotta figure from an archaeological excavation.', 'Terracotta', 'Telangana', '1000-01-01', 220000.00, '2004-03-18', 'Damaged
'),
    -> (19, 10, 10, 'ART-019', 'Ancient Gold Coin', 'Historical gold coin from a royal period.', 'Coin', 'Hyderabad', '1650-01-01', 750000.00, '2005-06-10', 'Good'),
    -> (20, 10, 10, 'ART-020', 'Silver Royal Coin', 'Historical silver coin belonging to a royal collection.', 'Coin', 'Hyderabad', '1700-01-01', 180000.00, '2006-08-25', 'Good');
Query OK, 20 rows affected (0.00 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO EXHIBITION (exhibition_id, museum_id, exhibition_name, description, theme, start_date, end_date, status) VALUES
    -> (1, 1, 'Indian Heritage Exhibition', 'Exhibition showcasing important Indian historical artifacts.', 'Indian Heritage', '2026-01-01', '2026-06-30', 'Active'),
    -> (2, 1, 'Ancient Sculptures', 'Collection of ancient Indian sculptures.', 'Ancient Art', '2026-03-01', '2026-08-31', 'Active'),
    -> (3, 2, 'Royal Mysuru Collection', 'Artifacts from the Mysuru royal period.', 'Royal Heritage', '2026-02-01', '2026-12-31', 'Active'),
    -> (4, 2, 'Palace Treasures', 'Royal furniture and ceremonial objects.', 'Royal Treasures', '2026-04-01', '2026-09-30', 'Active'),
    -> (5, 3, 'Masters of Indian Painting', 'Exhibition of famous Indian paintings.', 'Indian Painting', '2026-01-15', '2026-05-30', 'Completed'),
    -> (6, 4, 'Temple Heritage', 'Traditional temple artifacts from South India.', 'Temple Culture', '2026-05-01', '2026-10-31', 'Active'),
    -> (7, 4, 'Traditional Crafts', 'Exhibition of traditional South Indian crafts.', 'Traditional Crafts', '2026-06-01', '2026-11-30', 'Planned'),
    -> (8, 5, 'Archaeological Discoveries', 'Important archaeological discoveries.', 'Archaeology', '2026-02-15', '2026-07-31', 'Active');
Query OK, 8 rows affected (0.00 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO DISPLAY (exhibition_id, artifact_id, display_start_date, display_end_date, display_location, display_case, display_notes) VALUES
    -> (1, 1, '2026-01-01', '2026-06-30', 'Gallery A', 'A1', 'Ancient sculpture section'),
    -> (1, 3, '2026-01-01', '2026-06-30', 'Gallery A', 'A2', 'Historical manuscripts'),
    -> (1, 9, '2026-01-01', '2026-06-30', 'Gallery B', 'B1', 'Indian paintings'),
    -> (2, 2, '2026-03-01', '2026-08-31', 'Gallery C', 'C1', 'Granite sculpture'),
    -> (2, 14, '2026-03-01', '2026-08-31', 'Gallery C', 'C2', 'Temple stone panel'),
    -> (3, 5, '2026-02-01', '2026-12-31', 'Royal Gallery', 'R1', 'Royal furniture'),
    -> (3, 6, '2026-02-01', '2026-12-31', 'Royal Gallery', 'R2', 'Royal ceremonial sword'),
    -> (3, 7, '2026-02-01', '2026-12-31', 'Royal Gallery', 'R3', 'Royal throne'),
    -> (4, 8, '2026-04-01', '2026-09-30', 'Palace Gallery', 'P1', 'Royal jewellery box'),
    -> (5, 10, '2026-01-15', '2026-05-30', 'Painting Gallery', 'PG1', 'Bengal School painting'),
    -> (5, 11, '2026-01-15', '2026-05-30', 'Painting Gallery', 'PG2', 'Village life painting'),
    -> (6, 13, '2026-05-01', '2026-10-31', 'Temple Gallery', 'T1', 'Bronze idol'),
    -> (6, 14, '2026-05-01', '2026-10-31', 'Temple Gallery', 'T2', 'Temple stone panel'),
    -> (8, 17, '2026-02-15', '2026-07-31', 'Archaeology Gallery', 'AR1', 'Bronze vessel'),
    -> (8, 18, '2026-02-15', '2026-07-31', 'Archaeology Gallery', 'AR2', 'Terracotta figure');
Query OK, 15 rows affected (0.00 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO ARTIFACT_MEDIA (media_id, artifact_id, staff_id, media_type, file_name, file_url, description, file_size) VALUES
    -> (1, 1, 1, 'Image', 'artifact001.jpg', 'https://museum.com/media/artifact001.jpg', 'Front view of stone sculpture.', 250000),
    -> (2, 2, 1, 'Image', 'artifact002.jpg', 'https://museum.com/media/artifact002.jpg', 'Granite temple statue image.', 280000),
    -> (3, 3, 3, 'Document', 'artifact003.pdf', 'https://museum.com/media/artifact003.pdf', 'Digital copy of royal manuscript.', 850000),
    -> (4, 5, 1, 'Image', 'artifact005.jpg', 'https://museum.com/media/artifact005.jpg', 'Royal wooden chest image.', 310000),
    -> (5, 6, 1, 'Image', 'artifact006.jpg', 'https://museum.com/media/artifact006.jpg', 'Royal sword image.', 290000),
    -> (6, 7, 7, 'Image', 'artifact007.jpg', 'https://museum.com/media/artifact007.jpg', 'Royal throne image.', 350000),
    -> (7, 9, 1, 'Image', 'artifact009.jpg', 'https://museum.com/media/artifact009.jpg', 'Mythological painting photograph.', 420000),
    -> (8, 10, 7, 'Image', 'artifact010.jpg', 'https://museum.com/media/artifact010.jpg', 'Bharat Mata painting photograph.', 450000),
    -> (9, 13, 5, 'Image', 'artifact013.jpg', 'https://museum.com/media/artifact013.jpg', 'Temple bronze idol photograph.', 300000),
    -> (10, 15, 5, 'Image', 'artifact015.jpg', 'https://museum.com/media/artifact015.jpg', 'Traditional wooden doll photograph.', 180000),
    -> (11, 17, 6, 'Image', 'artifact017.jpg', 'https://museum.com/media/artifact017.jpg', 'Ancient bronze vessel photograph.', 320000),
    -> (12, 19, 6, 'Image', 'artifact019.jpg', 'https://museum.com/media/artifact019.jpg', 'Ancient gold coin photograph.', 150000);
Query OK, 12 rows affected (0.00 sec)
Records: 12  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO ARTIFACT_HISTORY (history_id, artifact_id, staff_id, event_date, event_type, location, description, source_reference) VALUES
    -> (1, 1, 1, '1986-03-12', 'Acquisition', 'National Heritage Museum', 'Artifact acquired and added to museum collection.', 'Acquisition Register 1986'),
    -> (2, 2, 1, '1987-04-15', 'Acquisition', 'National Heritage Museum', 'Granite statue added to collection.', 'Acquisition Register 1987'),
    -> (3, 3, 3, '1988-04-10', 'Acquisition', 'National Heritage Museum', 'Royal manuscript added to archives.', 'Archive Register 1988'),
    -> (4, 5, 1, '1994-02-15', 'Acquisition', 'Royal History Museum', 'Royal wooden chest acquired.', 'Royal Collection Register'),
    -> (5, 6, 1, '1995-07-11', 'Acquisition', 'Royal History Museum', 'Royal sword acquired.', 'Royal Collection Register'),
    -> (6, 7, 2, '1996-01-20', 'Acquisition', 'Royal History Museum', 'Royal throne added to collection.', 'Royal Collection Register'),
    -> (7, 9, 1, '1987-06-20', 'Acquisition', 'Indian Art Museum', 'Mythological painting acquired.', 'Painting Register'),
    -> (8, 10, 7, '1990-02-11', 'Acquisition', 'Indian Art Museum', 'Bharat Mata painting acquired.', 'Painting Register'),
    -> (9, 11, 7, '1995-03-22', 'Acquisition', 'Indian Art Museum', 'Village life painting acquired.', 'Painting Register'),
    -> (10, 13, 1, '1998-04-18', 'Acquisition', 'South Indian Heritage Museum', 'Bronze temple idol acquired.', 'Temple Artifact Register'),
    -> (11, 14, 1, '1999-08-20', 'Acquisition', 'South Indian Heritage Museum', 'Temple stone panel acquired.', 'Temple Artifact Register'),
    -> (12, 15, 6, '2001-05-10', 'Acquisition', 'South Indian Heritage Museum', 'Traditional wooden doll acquired.', 'Craft Register'),
    -> (13, 17, 6, '2003-01-22', 'Discovery', 'Ancient Civilisation Museum', 'Bronze vessel obtained through archaeological research.', 'Archaeology Report 2003'),
    -> (14, 18, 6, '2004-03-18', 'Discovery', 'Ancient Civilisation Museum', 'Terracotta figure discovered during excavation.', 'Archaeology Report 2004'),
    -> (15, 19, 6, '2005-06-10', 'Acquisition', 'Ancient Civilisation Museum', 'Ancient gold coin acquired.', 'Numismatic Register');
Query OK, 15 rows affected (0.00 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO RESTORATION_PROJECT (project_id, artifact_id, manager_id, project_name, objective, start_date, end_date, status, cost) VALUES
    -> (1, 9, 2, 'Mythological Painting Restoration', 'Restore damaged paint layers and improve structural stability.', '2026-01-15', '2026-05-15', 'In Progress', 75000.00),
    -> (2, 5, 2, 'Royal Wooden Chest Restoration', 'Repair structural damage and restore original surface finish.', '2026-02-01', '2026-07-01', 'Planned', 55000.00),
    -> (3, 7, 2, 'Royal Throne Conservation', 'Restore damaged wooden structure and decorative elements.', '2026-03-01', '2026-08-30', 'In Progress', 120000.00),
    -> (4, 13, 5, 'Bronze Idol Conservation', 'Clean and preserve the historical bronze surface.', '2026-04-01', '2026-06-30', 'Planned', 65000.00),
    -> (5, 16, 5, 'Textile Restoration', 'Restore damaged fibers and preserve traditional patterns.', '2026-05-01', '2026-09-30', 'Planned', 45000.00),
    -> (6, 18, 2, 'Terracotta Figure Restoration', 'Repair cracks and stabilize the terracotta artifact.', '2026-02-15', '2026-05-30', 'Completed', 38000.00),
    -> (7, 14, 5, 'Temple Stone Panel Conservation', 'Clean and stabilize the historical stone panel.', '2026-06-01', '2026-10-31', 'Planned', 70000.00),
    -> (8, 4, 2, 'Historical Document Preservation', 'Preserve and digitally archive the historical document.', '2026-01-10', '2026-04-30', 'Completed', 30000.00);
Query OK, 8 rows affected (0.00 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO CONDITION_ASSESSMENT (assessment_id, artifact_id, staff_id, assessment_date, conditions, damage_type, damage_description, recommended_action, priority) VALUES
    -> (1, 9, 2, '2026-01-10', 'Poor', 'Surface Damage', 'Paint layer is damaged in multiple areas.', 'Clean surface and restore damaged paint.', 'High'),
    -> (2, 5, 2, '2026-01-20', 'Fair', 'Structural Damage', 'Wooden structure has cracks and weakened joints.', 'Repair joints and treat damaged wood.', 'Medium'),
    -> (3, 7, 2, '2026-02-01', 'Poor', 'Structural Damage', 'Several wooden joints are unstable.', 'Structural restoration required.', 'High'),
    -> (4, 13, 5, '2026-02-10', 'Fair', 'Surface Corrosion', 'Bronze surface has developed oxidation.', 'Clean and apply conservation coating.', 'Medium'),
    -> (5, 16, 5, '2026-02-15', 'Poor', 'Fiber Damage', 'Several textile fibers are weakened.', 'Textile conservation required.', 'High'),
    -> (6, 18, 2, '2026-02-20', 'Poor', 'Cracking', 'Terracotta figure contains multiple cracks.', 'Stabilize and repair cracks.', 'High'),
    -> (7, 14, 5, '2026-03-05', 'Fair', 'Surface Wear', 'Stone surface shows signs of weathering.', 'Surface cleaning and stabilization.', 'Medium'),
    -> (8, 2, 1, '2026-03-10', 'Good', 'Minor Wear', 'Minor surface wear observed.', 'Routine monitoring.', 'Low'),
    -> (9, 3, 3, '2026-03-15', 'Good', 'Paper Aging', 'Paper has minor signs of aging.', 'Climate-controlled storage.', 'Low'),
    -> (10, 6, 1, '2026-03-20', 'Good', 'Minor Scratches', 'Minor scratches observed on surface.', 'Routine conservation.', 'Low'),
    -> (11, 10, 7, '2026-03-25', 'Good', 'Minor Wear', 'Minor frame and surface wear.', 'Routine conservation.', 'Low'),
    -> (12, 15, 5, '2026-04-01', 'Good', 'Surface Wear', 'Minor wear due to handling.', 'Preventive conservation.', 'Low');
Query OK, 12 rows affected (0.00 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> INSERT INTO RESTORATION_EXPERT (expert_id, expert_name, email, phone, hours_worked) VALUES
    -> (1, 'Dr. Arjun Kumar', 'arjun@restoration.com', '9988700001', 120.00),
    -> (2, 'Priya Menon', 'priya@restoration.com', '9988700002', 85.50),
    -> (3, 'Vikram Rao', 'vikram@restoration.com', '9988700003', 60.00),
    -> (4, 'Kavya Iyer', 'kavya@restoration.com', '9988700004', 95.00),
    -> (5, 'Suresh Bhat', 'suresh@restoration.com', '9988700005', 72.50),
    -> (6, 'Neha Sharma', 'neha@restoration.com', '9988700006', 110.00),
    -> (7, 'Rohan Das', 'rohan@restoration.com', '9988700007', 65.00),
    -> (8, 'Lakshmi Rao', 'lakshmi@restoration.com', '9988700008', 88.00);
Query OK, 8 rows affected (0.00 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO PROJECT_EXPERT (project_id, expert_id) VALUES
    -> (1, 1), (1, 2), (1, 4), (2, 2), (2, 3), (3, 1), (3, 5), (3, 6), (4, 4), (4, 7), (5, 6), (5, 8), (6, 3), (6, 5), (7, 1);
Query OK, 15 rows affected (0.00 sec)
Records: 15  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO RESTORATION_MATERIAL (material_id, material_name, material_type, supplier, unit, unit_cost, quantity_used, total_cost) VALUES
    -> (1, 'Archival Adhesive', 'Adhesive', 'Heritage Supplies', 'litre', 2500.00, 5.00, 12500.00),
    -> (2, 'Protective Varnish', 'Coating', 'Heritage Supplies', 'litre', 3500.00, 4.00, 14000.00),
    -> (3, 'Wood Treatment Oil', 'Wood Treatment', 'Conservation Materials Ltd', 'litre', 1800.00, 6.00, 10800.00),
    -> (4, 'Cleaning Solution', 'Cleaning', 'Conservation Materials Ltd', 'litre', 1200.00, 5.00, 6000.00),
    -> (5, 'Archival Paper', 'Paper', 'Museum Supplies India', 'sheet', 150.00, 100.00, 15000.00),
    -> (6, 'Protective Glass', 'Glass', 'Heritage Supplies', 'sqft', 850.00, 20.00, 17000.00),
    -> (7, 'Gold Leaf', 'Decorative', 'Traditional Materials Co', 'sheet', 450.00, 50.00, 22500.00),
    -> (8, 'Bronze Cleaner', 'Cleaning', 'Conservation Materials Ltd', 'litre', 2200.00, 4.00, 8800.00),
    -> (9, 'Textile Support Fabric', 'Textile', 'Heritage Textiles', 'metre', 900.00, 15.00, 13500.00),
    -> (10, 'Stone Consolidant', 'Stone Treatment', 'Heritage Supplies', 'litre', 3200.00, 5.00, 16000.00);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO PROJECT_MATERIAL (project_id, material_id) VALUES
    -> (1, 1), (1, 2), (1, 4), (2, 3), (2, 4), (2, 7), (3, 3), (3, 6), (3, 7), (4, 4), (4, 8), (5, 5), (5, 9), (6, 1), (6, 4), (7, 4), (7, 10),
(8, 1), (8, 5), (8, 6);
Query OK, 20 rows affected (0.00 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO VISITOR (visitor_id, visitor_name, email, phone, age, nationality, registration_date) VALUES
    -> (1, 'Aarav Singh', 'aarav@gmail.com', '9000000001', 24, 'Indian', '2026-01-05'),
    -> (2, 'Ananya Rao', 'ananya@gmail.com', '9000000002', 21, 'Indian', '2026-01-10'),
    -> (3, 'John Smith', 'john@gmail.com', '9000000003', 35, 'British', '2026-02-15'),
    -> (4, 'Sophia Lee', 'sophia@gmail.com', '9000000004', 28, 'Singaporean', '2026-03-01'),
    -> (5, 'Rahul Verma', 'rahulv@gmail.com', '9000000005', 30, 'Indian', '2026-03-05'),
    -> (6, 'Sneha Kapoor', 'sneha@gmail.com', '9000000006', 22, 'Indian', '2026-03-12'),
    -> (7, 'David Wilson', 'david@gmail.com', '9000000007', 42, 'British', '2026-03-20'),
    -> (8, 'Emily Brown', 'emily@gmail.com', '9000000008', 31, 'American', '2026-03-25'),
    -> (9, 'Vikram Shetty', 'vikram@gmail.com', '9000000009', 26, 'Indian', '2026-04-01'),
    -> (10, 'Priya Nair', 'priya@gmail.com', '9000000010', 23, 'Indian', '2026-04-05'),
    -> (11, 'Michael Chen', 'michael@gmail.com', '9000000011', 38, 'Singaporean', '2026-04-10'),
    -> (12, 'Kavya Menon', 'kavya@gmail.com', '9000000012', 27, 'Indian', '2026-04-15'),
    -> (13, 'Daniel Garcia', 'daniel@gmail.com', '9000000013', 34, 'Spanish', '2026-04-20'),
    -> (14, 'Meera Iyer', 'meera.visitor@gmail.com', '9000000014', 29, 'Indian', '2026-04-25'),
    -> (15, 'Lucas Martin', 'lucas@gmail.com', '9000000015', 40, 'French', '2026-05-01');
Query OK, 15 rows affected (0.00 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO VISIT_RECORD (visit_id, visitor_id, museum_id, visit_date, entry_time, exit_time, visit_type, ticket_type) VALUES
    -> (1, 1, 1, '2026-02-10', '10:00:00', '13:00:00', 'Individual', 'General'),
    -> (2, 2, 1, '2026-02-12', '11:00:00', '14:00:00', 'Individual', 'Student'),
    -> (3, 3, 2, '2026-03-05', '09:30:00', '12:30:00', 'Tourist', 'General'),
    -> (4, 4, 2, '2026-03-10', '14:00:00', '17:00:00', 'Tourist', 'General'),
    -> (5, 5, 1, '2026-03-15', '10:30:00', '13:30:00', 'Individual', 'General'),
    -> (6, 6, 3, '2026-03-18', '11:00:00', '15:00:00', 'Student', 'Student'),
    -> (7, 7, 3, '2026-03-20', '09:00:00', '12:00:00', 'Tourist', 'General'),
    -> (8, 8, 4, '2026-03-22', '10:00:00', '13:00:00', 'Tourist', 'General'),
    -> (9, 9, 4, '2026-03-25', '12:00:00', '15:00:00', 'Individual', 'Student'),
    -> (10, 10, 5, '2026-03-28', '09:30:00', '12:30:00', 'Tourist', 'General'),
    -> (11, 11, 5, '2026-04-01', '10:00:00', '14:00:00', 'Tourist', 'General'),
    -> (12, 12, 1, '2026-04-03', '11:00:00', '13:00:00', 'Individual', 'Student'),
    -> (13, 13, 2, '2026-04-05', '09:00:00', '12:00:00', 'Tourist', 'General'),
    -> (14, 14, 3, '2026-04-08', '13:00:00', '16:00:00', 'Individual', 'General'),
    -> (15, 15, 4, '2026-04-10', '10:30:00', '14:00:00', 'Tourist', 'General'),
    -> (16, 1, 5, '2026-04-12', '09:00:00', '12:00:00', 'Individual', 'General'),
    -> (17, 2, 2, '2026-04-15', '10:00:00', '13:00:00', 'Individual', 'Student'),
    -> (18, 3, 1, '2026-04-18', '11:00:00', '14:30:00', 'Tourist', 'General'),
    -> (19, 4, 3, '2026-04-20', '09:30:00', '12:30:00', 'Tourist', 'General'),
    -> (20, 5, 4, '2026-04-22', '10:00:00', '13:00:00', 'Individual', 'General'),
    -> (21, 6, 5, '2026-04-25', '12:00:00', '15:00:00', 'Student', 'Student'),
    -> (22, 7, 1, '2026-04-27', '09:00:00', '11:30:00', 'Tourist', 'General'),
    -> (23, 8, 2, '2026-04-29', '10:00:00', '13:00:00', 'Individual', 'General'),
    -> (24, 9, 3, '2026-05-01', '11:00:00', '14:00:00', 'Student', 'Student'),
    -> (25, 10, 5, '2026-05-03', '09:30:00', '12:30:00', 'Tourist', 'General');
Query OK, 25 rows affected (0.00 sec)
Records: 25  Duplicates: 0  Warnings: 0
```

Data modification and deletion:

```
mysql> UPDATE ARTIFACT SET estimated_value = 300000.00 WHERE artifact_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE ARTIFACT SET status = 'Needs Restoration' WHERE artifact_id = 2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM VISITOR WHERE visitor_id = 15;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select *from artifact;
+-------------+---------------+------------+------------------+-------------------------+-----------------------------------------------------------------+---------------------+--------------+---------------+-----------------+-------------------+
| artifact_id | collection_id | creator_id | accession_number | artifact_name           | description                                                     | artifact_type       | origin_place | creation_date | estimated_value | acquisition_date  | status            |
+-------------+---------------+------------+------------------+-------------------------+-----------------------------------------------------------------+---------------------+--------------+---------------+-----------------+-------------------+-------------------+
|           1 |             1 |          2 | ART-001          | Ancient Stone Sculpture | Stone sculpture depicting an ancient Indian deity.              | Sculpture           | Hampi        | 1250-01-01    |       300000.00 | 1986-03-12        | Good              |
|           2 |             1 |          2 | ART-002          | Granite Temple Statue   | Carved granite statue from a historic temple.                   | Sculpture           | Hampi        | 1300-01-01    |       320000.00 | 1987-04-15        | Needs Restoration |
|           3 |             2 |          4 | ART-003          | Royal Manuscript        | Handwritten manuscript containing historical records.           | Manuscript          | Mysuru       | 1650-01-01    |       150000.00 | 1988-04-10        | Good              |
|           4 |             2 |          4 | ART-004          | Historical Document     | Historical administrative document.                             | Document            | Karnataka    | 1700-01-01    |        80000.00 | 1990-08-19        | Good              |
|           5 |             3 |          3 | ART-005          | Royal Wooden Chest      | Decorated wooden chest used by a historical royal family.       | Furniture           | Mysuru       | 1780-01-01    |       180000.00 | 1994-02-15        | Damaged           |
|           6 |             3 |          3 | ART-006          | Royal Sword             | Ceremonial sword belonging to a royal collection.               | Weapon              | Mysuru       | 1775-01-01    |       300000.00 | 1995-07-11        | Good              |
|           7 |             4 |          3 | ART-007          | Royal Throne            | Decorated wooden throne used for ceremonial purposes.           | Furniture           | Mysuru       | 1800-01-01    |       550000.00 | 1996-01-20        | Needs Restoration |
|           8 |             4 |          3 | ART-008          | Royal Jewellery Box     | Ornamented jewellery box from the royal collection.             | Furniture           | Mysuru       | 1820-01-01    |       210000.00 | 1997-05-12        | Good              |
|           9 |             5 |          1 | ART-009          | Mythological Painting   | Painting depicting an Indian mythological scene.                | Painting            | Mysuru       | 1890-01-01    |       450000.00 | 1987-06-20        | Needs Restoration |
|          10 |             5 |          6 | ART-010          | Bharat Mata Painting    | Traditional Indian painting representing Bharat Mata.           | Painting            | Kolkata      | 1905-01-01    |       500000.00 | 1990-02-11        | Good              |
|          11 |             5 |          6 | ART-011          | Village Life Painting   | Painting depicting rural Indian life.                           | Painting            | Kolkata      | 1920-01-01    |       350000.00 | 1995-03-22        | Good              |
|          12 |             6 |          7 | ART-012          | Folk Art Composition    | Folk-inspired modern Indian artwork.                            | Painting            | West Bengal  | 1935-01-01    |       280000.00 | 2000-07-15        | Good              |
|          13 |             7 |          8 | ART-013          | Temple Bronze Idol      | Traditional bronze idol from a South Indian temple.             | Religious Artifact  | Tamil Nadu   | 1450-01-01    |       400000.00 | 1998-04-18        | Damaged           |
|          14 |             7 |          8 | ART-014          | Temple Stone Panel      | Decorative stone panel from an ancient temple.                  | Sculpture           | Tamil Nadu   | 1500-01-01    |       275000.00 | 1999-08-20        | Good              |
|          15 |             8 |          8 | ART-015          | Traditional Wooden Doll | Traditional South Indian wooden craft.                          | Craft               | Karnataka    | 1880-01-01    |        65000.00 | 2001-05-10        | Good              |
|          16 |             8 |          8 | ART-016          | Handwoven Textile       | Traditional handwoven textile with cultural patterns.           | Textile             | Tamil Nadu   | 1900-01-01    |        95000.00 | 2002-09-14        | Needs Restoration |
|          17 |             9 |          9 | ART-017          | Ancient Bronze Vessel   | Bronze vessel discovered during archaeological research.        | Vessel              | Andhra Pradesh | 1100-01-01  |       600000.00 | 2003-01-22        | Good              |
|          18 |             9 |          9 | ART-018          | Ancient Terracotta Figure | Terracotta figure from an archaeological excavation.          | Terracotta          | Telangana    | 1000-01-01    |       220000.00 | 2004-03-18        | Damaged           |
|          19 |            10 |         10 | ART-019          | Ancient Gold Coin       | Historical gold coin from a royal period.                       | Coin                | Hyderabad    | 1650-01-01    |       750000.00 | 2005-06-10        | Good              |
|          20 |            10 |         10 | ART-020          | Silver Royal Coin       | Historical silver coin belonging to a royal collection.         | Coin                | Hyderabad    | 1700-01-01    |       180000.00 | 2006-08-25        | Good              |
+-------------+---------------+------------+------------------+-------------------------+-----------------------------------------------------------------+---------------------+--------------+---------------+-----------------+-------------------+-------------------+
20 rows in set (0.00 sec)
```

```
mysql> select *from artifact_history;
+------------+-------------+----------+------------+--------------+-------------------------+-----------------------------------------------------------+-------------------------+
| history_id | artifact_id | staff_id | event_date | event_type   | location                | description                                               | source_reference        |
+------------+-------------+----------+------------+--------------+-------------------------+-----------------------------------------------------------+-------------------------+
|          1 |           1 |        1 | 1986-03-12 | Acquisition  | National Heritage Museum | Artifact acquired and added to museum collection.         | Acquisition Register 1986 |
|          2 |           2 |        1 | 1987-04-15 | Acquisition  | National Heritage Museum | Granite statue added to collection.                       | Acquisition Register 1987 |
|          3 |           3 |        3 | 1988-04-10 | Acquisition  | National Heritage Museum | Royal manuscript added to archives.                       | Archive Register 1988   |
|          4 |           5 |        1 | 1994-02-15 | Acquisition  | Royal History Museum     | Royal wooden chest acquired.                              | Royal Collection Register |
|          5 |           6 |        1 | 1995-07-11 | Acquisition  | Royal History Museum     | Royal sword acquired.                                     | Royal Collection Register |
|          6 |           7 |        2 | 1996-01-20 | Acquisition  | Royal History Museum     | Royal throne added to collection.                         | Royal Collection Register |
|          7 |           9 |        1 | 1987-06-20 | Acquisition  | Indian Art Museum        | Mythological painting acquired.                           | Painting Register       |
|          8 |          10 |        7 | 1990-02-11 | Acquisition  | Indian Art Museum        | Bharat Mata painting acquired.                            | Painting Register       |
|          9 |          11 |        7 | 1995-03-22 | Acquisition  | Indian Art Museum        | Village life painting acquired.                           | Painting Register       |
|         10 |          13 |        1 | 1998-04-18 | Acquisition  | South Indian Heritage Museum | Bronze temple idol acquired.                          | Temple Artifact Register |
|         11 |          14 |        1 | 1999-08-20 | Acquisition  | South Indian Heritage Museum | Temple stone panel acquired.                          | Temple Artifact Register |
|         12 |          15 |        6 | 2001-05-10 | Acquisition  | South Indian Heritage Museum | Traditional wooden doll acquired.                     | Craft Register          |
|         13 |          17 |        6 | 2003-01-22 | Discovery    | Ancient Civilisation Museum | Bronze vessel obtained through archaeological research. | Archaeology Report 2003 |
|         14 |          18 |        6 | 2004-03-18 | Discovery    | Ancient Civilisation Museum | Terracotta figure discovered during excavation.       | Archaeology Report 2004 |
|         15 |          19 |        6 | 2005-06-10 | Acquisition  | Ancient Civilisation Museum | Ancient gold coin acquired.                           | Numismatic Register     |
+------------+-------------+----------+------------+--------------+-------------------------+-----------------------------------------------------------+-------------------------+
15 rows in set (0.00 sec)

mysql> select *from artifact_media;
+----------+-------------+----------+------------+-----------+-----------------------------------------+-----------------------------------+---------------------+-----------+
| media_id | artifact_id | staff_id | media_type | file_name | file_url                                | description                       | uploaded_on         | file_size |
+----------+-------------+----------+------------+-----------+-----------------------------------------+-----------------------------------+---------------------+-----------+
|        1 |           1 |        1 | Image      | artifact001.jpg | https://museum.com/media/artifact001.jpg | Front view of stone sculpture.    | 2026-11-02 19:27:24 |    250000 |
|        2 |           2 |        1 | Image      | artifact002.jpg | https://museum.com/media/artifact002.jpg | Granite temple statue image.      | 2026-11-02 19:27:24 |    280000 |
|        3 |           3 |        3 | Document   | artifact003.pdf | https://museum.com/media/artifact003.pdf | Digital copy of royal manuscript. | 2026-11-02 19:27:24 |    850000 |
|        4 |           5 |        1 | Image      | artifact005.jpg | https://museum.com/media/artifact005.jpg | Royal wooden chest image.         | 2026-11-02 19:27:24 |    310000 |
|        5 |           6 |        1 | Image      | artifact006.jpg | https://museum.com/media/artifact006.jpg | Royal sword image.                | 2026-11-02 19:27:24 |    290000 |
|        6 |           7 |        7 | Image      | artifact007.jpg | https://museum.com/media/artifact007.jpg | Royal throne image.               | 2026-11-02 19:27:24 |    350000 |
|        7 |           9 |        1 | Image      | artifact009.jpg | https://museum.com/media/artifact009.jpg | Mythological painting photograph. | 2026-11-02 19:27:24 |    420000 |
|        8 |          10 |        7 | Image      | artifact010.jpg | https://museum.com/media/artifact010.jpg | Bharat Mata painting photograph.  | 2026-11-02 19:27:24 |    450000 |
|        9 |          13 |        5 | Image      | artifact013.jpg | https://museum.com/media/artifact013.jpg | Temple bronze idol photograph.    | 2026-11-02 19:27:24 |    300000 |
|       10 |          15 |        5 | Image      | artifact015.jpg | https://museum.com/media/artifact015.jpg | Traditional wooden doll photograph. | 2026-11-02 19:27:24 |    180000 |
|       11 |          17 |        6 | Image      | artifact017.jpg | https://museum.com/media/artifact017.jpg | Ancient bronze vessel photograph. | 2026-11-02 19:27:24 |    320000 |
|       12 |          19 |        6 | Image      | artifact019.jpg | https://museum.com/media/artifact019.jpg | Ancient gold coin photograph.     | 2026-11-02 19:27:24 |    150000 |
+----------+-------------+----------+------------+-----------+-----------------------------------------+-----------------------------------+---------------------+-----------+
12 rows in set (0.00 sec)
```

```
mysql> select *from collection;
+---------------+-----------+-------------------------+-----------------------------------------------------+-----------------+-----------------+--------------+
| collection_id | museum_id | collection_name         | description                                         | historical_period | collection_type | date_created |
+---------------+-----------+-------------------------+-----------------------------------------------------+-----------------+-----------------+--------------+
|             1 |         1 | Ancient Indian Art      | Collection of ancient Indian sculptures and artwork. | Ancient India   | Sculpture       | 1986-01-10   |
|             2 |         1 | Medieval Manuscripts    | Collection of historical manuscripts and documents. | Medieval Period | Manuscript      | 1987-05-15   |
|             3 |         2 | Royal Heritage          | Artifacts belonging to historical royal families.   | Medieval India  | Royal Artifacts | 1993-03-20   |
|             4 |         2 | Mysuru Palace Collection | Objects associated with Mysuru royal history.      | 18th-20th Century | Royal Collection | 1994-04-15 |
|             5 |         3 | Indian Paintings        | Collection of traditional Indian paintings.         | 19th-20th Century | Painting        | 1979-02-12   |
|             6 |         3 | Modern Art              | Collection of modern Indian artworks.               | 20th-21st Century | Modern Art      | 1985-07-21   |
|             7 |         4 | Temple Artifacts        | Artifacts collected from South Indian temples.      | Ancient-Medieval | Religious Art   | 1996-03-10   |
|             8 |         4 | Traditional Crafts      | Traditional crafts and handmade cultural objects.   | 19th-20th Century | Craft          | 1997-08-18   |
|             9 |         5 | Archaeological Collection | Artifacts discovered through archaeological research. | Ancient Period | Archaeological | 2002-01-20 |
|            10 |         5 | Coins and Currency      | Historical coins and currency from different periods. | Ancient-Modern | Numismatics     | 2003-06-11   |
+---------------+-----------+-------------------------+-----------------------------------------------------+-----------------+-----------------+--------------+
10 rows in set (0.00 sec)

mysql> select *from condition_assessment;
+---------------+-------------+----------+-----------------+------------+-----------------+-----------------------------------------+-----------------------------------------+----------+
| assessment_id | artifact_id | staff_id | assessment_date | conditions | damage_type     | damage_description                      | recommended_action                      | priority |
+---------------+-------------+----------+-----------------+------------+-----------------+-----------------------------------------+-----------------------------------------+----------+
|             1 |           9 |        2 | 2026-01-10      | Poor       | Surface Damage  | Paint layer is damaged in multiple areas. | Clean surface and restore damaged paint. | High     |
|             2 |           5 |        2 | 2026-01-20      | Fair       | Structural Damage | Wooden structure has cracks and weakened joints. | Repair joints and treat damaged wood. | Medium   |
|             3 |           7 |        2 | 2026-02-01      | Poor       | Structural Damage | Several wooden joints are unstable.     | Structural restoration required.        | High     |
|             4 |          13 |        5 | 2026-02-10      | Fair       | Surface Corrosion | Bronze surface has developed oxidation. | Clean and apply conservation coating.   | Medium   |
|             5 |          16 |        5 | 2026-02-15      | Poor       | Fiber Damage    | Several textile fibers are weakened.    | Textile conservation required.          | High     |
|             6 |          18 |        2 | 2026-02-20      | Poor       | Cracking        | Terracotta figure contains multiple cracks. | Stabilize and repair cracks.        | High     |
|             7 |          14 |        5 | 2026-03-05      | Fair       | Surface Wear    | Stone surface shows signs of weathering. | Surface cleaning and stabilization.    | Medium   |
|             8 |           2 |        1 | 2026-03-10      | Good       | Minor Wear      | Minor surface wear observed.            | Routine monitoring.                     | Low      |
|             9 |           3 |        3 | 2026-03-15      | Good       | Paper Aging     | Paper has minor signs of aging.         | Climate-controlled storage.             | Low      |
|            10 |           6 |        1 | 2026-03-20      | Good       | Minor Scratches | Minor scratches observed on surface.    | Routine conservation.                   | Low      |
|            11 |          10 |        7 | 2026-03-25      | Good       | Minor Wear      | Minor frame and surface wear.           | Routine conservation.                   | Low      |
|            12 |          15 |        5 | 2026-04-01      | Good       | Surface Wear    | Minor wear due to handling.             | Preventive conservation.                | Low      |
+---------------+-------------+----------+-----------------+------------+-----------------+-----------------------------------------+-----------------------------------------+----------+
12 rows in set (0.00 sec)
```

```
mysql> select *from creator
    -> ;
+------------+-------------------------+------------+-------------+-----------------------------------------------------------------+
| creator_id | creator_name            | birth_year | nationality | biography                                                       |
+------------+-------------------------+------------+-------------+-----------------------------------------------------------------+
|          1 | Ravi Varma              |       1848 | Indian      | Indian painter known for paintings based on Indian mythology.   |
|          2 | Unknown Ancient Sculptor|       1200 | Indian      | Ancient sculptor whose identity remains unknown.                |
|          3 | Mysuru Royal Artisan    |       1750 | Indian      | Traditional artisan associated with the Mysuru royal court.     |
|          4 | Unknown Manuscript Artist|      1600 | Indian      | Artist responsible for illustrating historical manuscripts.     |
|          5 | Abanindranath Tagore    |       1871 | Indian      | Indian painter and one of the founders of the Bengal School of Art. |
|          6 | Nandalal Bose           |       1882 | Indian      | Indian painter associated with the Bengal School of Art.        |
|          7 | Jamini Roy              |       1887 | Indian      | Indian painter known for folk-inspired artistic styles.         |
|          8 | Unknown Temple Artisan  |       1400 | Indian      | Traditional artisan responsible for temple artwork.             |
|          9 | Ancient Metalworker     |       1100 | Indian      | Ancient artisan specializing in metal objects.                  |
|         10 | Royal Coin Maker        |       1650 | Indian      | Artisan associated with historical royal coin production.       |
+------------+-------------------------+------------+-------------+-----------------------------------------------------------------+
10 rows in set (0.00 sec)

mysql> select *from display;
+---------------+-------------+--------------------+------------------+------------------+--------------+-------------------------+
| exhibition_id | artifact_id | display_start_date | display_end_date | display_location | display_case | display_notes           |
+---------------+-------------+--------------------+------------------+------------------+--------------+-------------------------+
|             1 |           1 | 2026-01-01         | 2026-06-30       | Gallery A        | A1           | Ancient sculpture section |
|             1 |           3 | 2026-01-01         | 2026-06-30       | Gallery A        | A2           | Historical manuscripts  |
|             1 |           9 | 2026-01-01         | 2026-06-30       | Gallery B        | B1           | Indian paintings        |
|             2 |           2 | 2026-03-01         | 2026-08-31       | Gallery C        | C1           | Granite sculpture       |
|             2 |          14 | 2026-03-01         | 2026-08-31       | Gallery C        | C2           | Temple stone panel      |
|             3 |           5 | 2026-02-01         | 2026-12-31       | Royal Gallery    | R1           | Royal furniture         |
|             3 |           6 | 2026-02-01         | 2026-12-31       | Royal Gallery    | R2           | Royal ceremonial sword  |
|             3 |           7 | 2026-02-01         | 2026-12-31       | Royal Gallery    | R3           | Royal throne            |
|             4 |           8 | 2026-04-01         | 2026-09-30       | Palace Gallery   | P1           | Royal jewellery box     |
|             5 |          10 | 2026-01-15         | 2026-05-30       | Painting Gallery | PG1          | Bengal School painting  |
|             5 |          11 | 2026-01-15         | 2026-05-30       | Painting Gallery | PG2          | Village life painting   |
|             6 |          13 | 2026-05-01         | 2026-10-31       | Temple Gallery   | T1           | Bronze idol             |
|             6 |          14 | 2026-05-01         | 2026-10-31       | Temple Gallery   | T2           | Temple stone panel      |
|             8 |          17 | 2026-02-15         | 2026-07-31       | Archaeology Gallery | AR1       | Bronze vessel           |
|             8 |          18 | 2026-02-15         | 2026-07-31       | Archaeology Gallery | AR2       | Terracotta figure       |
+---------------+-------------+--------------------+------------------+------------------+--------------+-------------------------+
15 rows in set (0.00 sec)
```

```
mysql> select *from exhibition;
+---------------+------------+-------------------------+-----------------------------------------------------------+-------------------+------------+------------+-----------+
| exhibition_id | museum_id  | exhibition_name         | description                                               | theme             | start_date | end_date   | status    |
+---------------+------------+-------------------------+-----------------------------------------------------------+-------------------+------------+------------+-----------+
|             1 |          1 | Indian Heritage Exhibition | Exhibition showcasing important Indian historical artifacts. | Indian Heritage | 2026-01-01 | 2026-06-30 | Active    |
|             2 |          1 | Ancient Sculptures      | Collection of ancient Indian sculptures.                  | Ancient Art       | 2026-03-01 | 2026-08-31 | Active    |
|             3 |          2 | Royal Mysuru Collection | Artifacts from the Mysuru royal period.                   | Royal Heritage    | 2026-02-01 | 2026-12-31 | Active    |
|             4 |          2 | Palace Treasures        | Royal furniture and ceremonial objects.                   | Royal Treasures   | 2026-04-01 | 2026-09-30 | Active    |
|             5 |          3 | Masters of Indian Painting | Exhibition of famous Indian paintings.                 | Indian Painting   | 2026-01-15 | 2026-05-30 | Completed |
|             6 |          4 | Temple Heritage         | Traditional temple artifacts from South India.            | Temple Culture    | 2026-05-01 | 2026-10-31 | Active    |
|             7 |          4 | Traditional Crafts      | Exhibition of traditional South Indian crafts.            | Traditional Crafts | 2026-06-01 | 2026-11-30 | Planned   |
|             8 |          5 | Archaeological Discoveries | Important archaeological discoveries.                  | Archaeology       | 2026-02-15 | 2026-07-31 | Active    |
+---------------+------------+-------------------------+-----------------------------------------------------------+-------------------+------------+------------+-----------+
8 rows in set (0.00 sec)
```

```
mysql> select *from museum;
+-----------+---------------------+-----------------------------------------------------------+---------------+-----------+-----------+---------+-----------------------+---------------+------------------+
| museum_id | museum_name         | description                                               | address       | city      | state     | country | contact_email         | contact_phone | established_year |
+-----------+---------------------+-----------------------------------------------------------+---------------+-----------+-----------+---------+-----------------------+---------------+------------------+
|         1 | National Heritage Museum | Museum preserving Indian historical and cultural artifacts. | MG Road | Bengaluru | Karnataka | India | info@nationalheritage.com | 08040001001 |             1985 |
|         2 | Royal History Museum | Museum dedicated to royal history and traditional artifacts. | Palace Road | Mysuru | Karnataka | India | info@royalhistory.com | 08214000102 |             1992 |
|         3 | Indian Art Museum   | Museum showcasing traditional and modern Indian art.      | Museum Road   | New Delhi | Delhi     | India | info@indianartmuseum.com | 01140001003 |             1978 |
|         4 | South Indian Heritage Museum | Museum preserving cultural heritage from South India. | Temple Street | Chennai | Tamil Nadu | India | info@sihmuseum.com | 04440001004 |             1995 |
|         5 | Ancient Civilisation Museum | Museum containing archaeological and ancient civilisation artifacts. | Heritage Avenue | Hyderabad | Telangana | India | info@ancientcivilisation.com | 04040001005 |             2001 |
+-----------+---------------------+-----------------------------------------------------------+---------------+-----------+-----------+---------+-----------------------+---------------+------------------+
5 rows in set (0.00 sec)
```

```
mysql> select *from project_expert;
+------------+------------+
| project_id | expert_id  |
+------------+------------+
|          1 |          1 |
|          3 |          1 |
|          7 |          1 |
|          1 |          2 |
|          2 |          2 |
|          2 |          3 |
|          6 |          3 |
|          1 |          4 |
|          4 |          4 |
|          3 |          5 |
|          6 |          5 |
|          3 |          6 |
|          5 |          6 |
|          4 |          7 |
|          5 |          8 |
+------------+------------+
15 rows in set (0.00 sec)
```

```
mysql> select *from project_material;
+------------+-------------+
| project_id | material_id |
+------------+-------------+
|          1 |           1 |
|          6 |           1 |
|          8 |           1 |
|          1 |           2 |
|          2 |           3 |
|          3 |           3 |
|          1 |           4 |
|          2 |           4 |
|          4 |           4 |
|          6 |           4 |
|          7 |           4 |
|          5 |           5 |
|          8 |           5 |
|          3 |           6 |
|          8 |           6 |
|          2 |           7 |
|          3 |           7 |
|          4 |           8 |
|          5 |           9 |
|          7 |          10 |
+------------+-------------+
20 rows in set (0.00 sec)
```

```
mysql> select *from restoration_expert;
+------------+-----------------+--------------------------+-------------+--------------+
| expert_id  | expert_name     | email                    | phone       | hours_worked |
+------------+-----------------+--------------------------+-------------+--------------+
|          1 | Dr. Arjun Kumar | arjun@restoration.com    | 9988700001  |       120.00 |
|          2 | Priya Menon     | priya@restoration.com    | 9988700002  |        85.50 |
|          3 | Vikram Rao      | vikram@restoration.com   | 9988700003  |        60.00 |
|          4 | Kavya Iyer      | kavya@restoration.com    | 9988700004  |        95.00 |
|          5 | Suresh Bhat     | suresh@restoration.com   | 9988700005  |        72.50 |
|          6 | Neha Sharma     | neha@restoration.com     | 9988700006  |       110.00 |
|          7 | Rohan Das       | rohan@restoration.com    | 9988700007  |        65.00 |
|          8 | Lakshmi Rao     | lakshmi@restoration.com  | 9988700008  |        88.00 |
+------------+-----------------+--------------------------+-------------+--------------+
8 rows in set (0.00 sec)
```

```
mysql> select *from restoration_material;
+-------------+-------------------------+-----------------+-----------------------------+-------+-----------+---------------+-------------+
| material_id | material_name           | material_type   | supplier                    | unit  | unit_cost | quantity_used | total_cost  |
+-------------+-------------------------+-----------------+-----------------------------+-------+-----------+---------------+-------------+
|           1 | Archival Adhesive       | Adhesive        | Heritage Supplies           | litre |   2500.00 |          5.00 |    12500.00 |
|           2 | Protective Varnish      | Coating         | Heritage Supplies           | litre |   3500.00 |          4.00 |    14000.00 |
|           3 | Wood Treatment Oil      | Wood Treatment  | Conservation Materials Ltd  | litre |   1800.00 |          6.00 |    10800.00 |
|           4 | Cleaning Solution       | Cleaning        | Conservation Materials Ltd  | litre |   1200.00 |          5.00 |     6000.00 |
|           5 | Archival Paper          | Paper           | Museum Supplies India       | sheet |    150.00 |        100.00 |    15000.00 |
|           6 | Protective Glass        | Glass           | Heritage Supplies           | sqft  |    850.00 |         20.00 |    17000.00 |
|           7 | Gold Leaf               | Decorative      | Traditional Materials Co    | sheet |    450.00 |         50.00 |    22500.00 |
|           8 | Bronze Cleaner          | Cleaning        | Conservation Materials Ltd  | litre |   2200.00 |          4.00 |     8800.00 |
|           9 | Textile Support Fabric  | Textile         | Heritage Textiles           | metre |    900.00 |         15.00 |    13500.00 |
|          10 | Stone Consolidant       | Stone Treatment | Heritage Supplies           | litre |   3200.00 |          5.00 |    16000.00 |
+-------------+-------------------------+-----------------+-----------------------------+-------+-----------+---------------+-------------+
10 rows in set (0.00 sec)
```

```
mysql> select *from restoration_project;
+------------+-------------+------------+-------------------------------+-----------------------------------------------------------+------------+------------+-------------+-----------+
| project_id | artifact_id | manager_id | project_name                  | objective                                                 | start_date | end_date   | status      | cost      |
+------------+-------------+------------+-------------------------------+-----------------------------------------------------------+------------+------------+-------------+-----------+
|          1 |           9 |          2 | Mythological Painting Restoration | Restore damaged paint layers and improve structural stability. | 2026-01-15 | 2026-05-15 | In Progress |  75000.00 |
|          2 |           5 |          2 | Royal Wooden Chest Restoration | Repair structural damage and restore original surface finish. | 2026-02-01 | 2026-07-01 | Planned     |  55000.00 |
|          3 |           7 |          2 | Royal Throne Conservation     | Restore damaged wooden structure and decorative elements. | 2026-03-01 | 2026-08-30 | In Progress | 120000.00 |
|          4 |          13 |          5 | Bronze Idol Conservation      | Clean and preserve the historical bronze surface.         | 2026-04-01 | 2026-06-30 | Planned     |  65000.00 |
|          5 |          16 |          5 | Textile Restoration           | Restore damaged fibers and preserve traditional patterns. | 2026-05-01 | 2026-09-30 | Planned     |  45000.00 |
|          6 |          18 |          2 | Terracotta Figure Restoration | Repair cracks and stabilize the terracotta artifact.      | 2026-02-15 | 2026-05-30 | Completed   |  38000.00 |
|          7 |          14 |          5 | Temple Stone Panel Conservation | Clean and stabilize the historical stone panel.         | 2026-06-01 | 2026-10-31 | Planned     |  70000.00 |
|          8 |           4 |          2 | Historical Document Preservation | Preserve and digitally archive the historical document. | 2026-01-10 | 2026-04-30 | Completed   |  30000.00 |
+------------+-------------+------------+-------------------------------+-----------------------------------------------------------+------------+------------+-------------+-----------+
8 rows in set (0.00 sec)
```

```
mysql> select *from staff;
+----------+-------------+--------------------+------------+--------------------+-------------------+---------------+---------------+--------------+-------+
| staff_id | staff_name  | email              | phone      | designation        | department        | username      | password_hash | staff_status | role  |
+----------+-------------+--------------------+------------+--------------------+-------------------+---------------+---------------+--------------+-------+
|        1 | Anita Rao   | anita@museum.com   | 9876500001 | Curator            | Collections       | anita.rao     | hash_anita123 | Active       | STAFF |
|        2 | Rahul Sharma | rahul@museum.com  | 9876500002 | Restoration Manager | Restoration      | rahul.sharma  | hash_rahul123 | Active       | STAFF |
|        3 | Meera Nair  | meera@museum.com   | 9876500003 | Archivist          | Archives          | meera.nair    | hash_meera123 | Active       | STAFF |
|        4 | Kiran Kumar | kiran@museum.com   | 9876500004 | Museum Director    | Administration    | kiran.kumar   | hash_kiran123 | Active       | STAFF |
|        5 | Priya Menon | priya@museum.com   | 9876500005 | Conservator        | Restoration       | priya.menon   | hash_priya123 | Active       | STAFF |
|        6 | Vivek Rao   | vivek@museum.com   | 9876500006 | Researcher         | Research          | vivek.rao     | hash_vivek123 | Active       | STAFF |
|        7 | Sneha Iyer  | sneha@museum.com   | 9876500007 | Curator            | Exhibitions       | sneha.iyer    | hash_sneha123 | Active       | STAFF |
|        8 | Arjun Das   | arjun@museum.com   | 9876500008 | Security Officer   | Security          | arjun.das     | hash_arjun123 | Active       | STAFF |
|        9 | Divya Shah  | divya@museum.com   | 9876500009 | Archivist          | Archives          | divya.shah    | hash_divya123 | Active       | STAFF |
|       10 | Manoj Kumar | manoj@museum.com   | 9876500010 | Technician         | Digital Archives  | manoj.kumar   | hash_manoj123 | Active       | STAFF |
+----------+-------------+--------------------+------------+--------------------+-------------------+---------------+---------------+--------------+-------+
10 rows in set (0.00 sec)
```

```
mysql> select *from visit_record;
+----------+------------+-----------+------------+------------+-----------+--------------+-------------+
| visit_id | visitor_id | museum_id | visit_date | entry_time | exit_time | visit_type   | ticket_type |
+----------+------------+-----------+------------+------------+-----------+--------------+-------------+
|        1 |          1 |         1 | 2026-02-10 | 10:00:00   | 13:00:00  | Individual   | General     |
|        2 |          2 |         1 | 2026-02-12 | 11:00:00   | 14:00:00  | Individual   | Student     |
|        3 |          3 |         2 | 2026-03-05 | 09:30:00   | 12:30:00  | Tourist      | General     |
|        4 |          4 |         2 | 2026-03-10 | 14:00:00   | 17:00:00  | Tourist      | General     |
|        5 |          5 |         1 | 2026-03-15 | 10:30:00   | 13:30:00  | Individual   | General     |
|        6 |          6 |         3 | 2026-03-18 | 11:00:00   | 15:00:00  | Student      | Student     |
|        7 |          7 |         3 | 2026-03-20 | 09:00:00   | 12:00:00  | Tourist      | General     |
|        8 |          8 |         4 | 2026-03-22 | 10:00:00   | 13:00:00  | Tourist      | General     |
|        9 |          9 |         4 | 2026-03-25 | 12:00:00   | 15:00:00  | Individual   | Student     |
|       10 |         10 |         5 | 2026-03-28 | 09:30:00   | 12:30:00  | Tourist      | General     |
|       11 |         11 |         5 | 2026-04-01 | 10:00:00   | 14:00:00  | Tourist      | General     |
|       12 |         12 |         1 | 2026-04-03 | 11:00:00   | 13:00:00  | Individual   | Student     |
|       13 |         13 |         2 | 2026-04-05 | 09:00:00   | 12:00:00  | Tourist      | General     |
|       14 |         14 |         3 | 2026-04-08 | 13:00:00   | 16:00:00  | Individual   | General     |
|       16 |          1 |         5 | 2026-04-12 | 09:00:00   | 12:00:00  | Individual   | General     |
|       17 |          2 |         2 | 2026-04-15 | 10:00:00   | 13:00:00  | Individual   | Student     |
|       18 |          3 |         1 | 2026-04-18 | 11:00:00   | 14:30:00  | Tourist      | General     |
|       19 |          4 |         3 | 2026-04-20 | 09:30:00   | 12:30:00  | Tourist      | General     |
|       20 |          5 |         4 | 2026-04-22 | 10:00:00   | 13:00:00  | Individual   | General     |
|       21 |          6 |         5 | 2026-04-25 | 12:00:00   | 15:00:00  | Student      | Student     |
|       22 |          7 |         1 | 2026-04-27 | 09:00:00   | 11:30:00  | Tourist      | General     |
|       23 |          8 |         2 | 2026-04-29 | 10:00:00   | 13:00:00  | Individual   | General     |
|       24 |          9 |         3 | 2026-05-01 | 11:00:00   | 14:00:00  | Student      | Student     |
|       25 |         10 |         5 | 2026-05-03 | 09:30:00   | 12:30:00  | Tourist      | General     |
+----------+------------+-----------+------------+------------+-----------+--------------+-------------+
24 rows in set (0.00 sec)

mysql> select *from visitor;
+------------+---------------+-----------------------+------------+-----+-------------+-------------------+
| visitor_id | visitor_name  | email                 | phone      | age | nationality | registration_date |
+------------+---------------+-----------------------+------------+-----+-------------+-------------------+
|          1 | Aarav Singh   | aarav@gmail.com       | 9000000001 |  24 | Indian      | 2026-01-05        |
|          2 | Ananya Rao    | ananya@gmail.com      | 9000000002 |  21 | Indian      | 2026-01-10        |
|          3 | John Smith    | john@gmail.com        | 9000000003 |  35 | British     | 2026-02-15        |
|          4 | Sophia Lee    | sophia@gmail.com      | 9000000004 |  28 | Singaporean | 2026-03-01        |
|          5 | Rahul Verma   | rahulv@gmail.com      | 9000000005 |  30 | Indian      | 2026-03-05        |
|          6 | Sneha Kapoor  | sneha@gmail.com       | 9000000006 |  22 | Indian      | 2026-03-12        |
|          7 | David Wilson  | david@gmail.com       | 9000000007 |  42 | British     | 2026-03-20        |
|          8 | Emily Brown   | emily@gmail.com       | 9000000008 |  31 | American    | 2026-03-25        |
|          9 | Vikram Shetty | vikram@gmail.com      | 9000000009 |  26 | Indian      | 2026-04-01        |
|         10 | Priya Nair    | priya@gmail.com       | 9000000010 |  23 | Indian      | 2026-04-05        |
|         11 | Michael Chen  | michael@gmail.com     | 9000000011 |  38 | Singaporean | 2026-04-10        |
|         12 | Kavya Menon   | kavya@gmail.com       | 9000000012 |  27 | Indian      | 2026-04-15        |
|         13 | Daniel Garcia | daniel@gmail.com      | 9000000013 |  34 | Spanish     | 2026-04-20        |
|         14 | Meera Iyer    | meera.visitor@gmail.com | 9000000014 |  29 | Indian      | 2026-04-25        |
+------------+---------------+-----------------------+------------+-----+-------------+-------------------+
14 rows in set (0.00 sec)

mysql>
```

Data integrity

Foreign keys prevent records from referencing missing parent rows. NOT NULL constraints protect required values. UNIQUE constraints prevent duplicate accession numbers, staff email addresses, and usernames. Associative-table composite keys prevent duplicate links between the same pair of records.

7. QUERIES

7.1 SIMPLE QUERY WITH GROUP BY, AGRREGATE

```
mysql> SELECT artifact_type, COUNT(*) AS total_artifacts
    -> FROM ARTIFACT
    -> GROUP BY artifact_type
    -> ORDER BY total_artifacts DESC;
+--------------------+-----------------+
| artifact_type      | total_artifacts |
+--------------------+-----------------+
| Painting           |               4 |
| Sculpture          |               3 |
| Furniture          |               3 |
| Coin               |               2 |
| Manuscript         |               1 |
| Document           |               1 |
| Weapon             |               1 |
| Religious Artifact |               1 |
| Craft              |               1 |
| Textile            |               1 |
| Vessel             |               1 |
| Terracotta         |               1 |
+--------------------+-----------------+
12 rows in set (0.00 sec)
```

This query groups artifacts by type and counts the number of artifacts in each group. COUNT(*) is an aggregate function and GROUP BY creates one result row per artifact type.

```
mysql> SELECT COUNT(*) AS total_artifacts,
    ->        AVG(estimated_value) AS average_value,
    ->        MIN(estimated_value) AS min_value,
    ->        MAX(estimated_value) AS max_value,
    ->        SUM(estimated_value) AS total_value
    -> FROM ARTIFACT
    -> ;
+-----------------+---------------+-----------+-----------+-------------+
| total_artifacts | average_value | min_value | max_value | total_value |
+-----------------+---------------+-----------+-----------+-------------+
|              20 | 312750.000000 |  65000.00 | 750000.00 |  6255000.00 |
+-----------------+---------------+-----------+-----------+-------------+
1 row in set (0.00 sec)
```

This query demonstrates COUNT, AVG, MIN, MAX, and SUM over the artifact catalogue.

7.2 UPDATE OPERATION

```
mysql> UPDATE ARTIFACT
    -> SET status = 'Needs Restoration'
    -> WHERE artifact_id = 2;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0
```

The UPDATE statement changes the status of the specified artifact. The WHERE clause ensures that only the row matching artifact_id 2 is affected.

7.3 DELETE OPERATION

```
mysql> DELETE FROM VISITOR
    -> WHERE visitor_id = 15;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from visitor;
+------------+--------------+-------------------------+------------+------+-------------+-------------------+
| visitor_id | visitor_name | email                   | phone      | age  | nationality | registration_date |
+------------+--------------+-------------------------+------------+------+-------------+-------------------+
|          1 | Aarav Singh  | aarav@gmail.com         | 9000000001 |   24 | Indian      | 2026-01-05        |
|          2 | Ananya Rao   | ananya@gmail.com        | 9000000002 |   21 | Indian      | 2026-01-10        |
|          3 | John Smith   | john@gmail.com          | 9000000003 |   35 | British     | 2026-02-15        |
|          4 | Sophia Lee   | sophia@gmail.com        | 9000000004 |   28 | Singaporean | 2026-03-01        |
|          5 | Rahul Verma  | rahulv@gmail.com        | 9000000005 |   30 | Indian      | 2026-03-05        |
|          6 | Sneha Kapoor | sneha@gmail.com         | 9000000006 |   22 | Indian      | 2026-03-12        |
|          7 | David Wilson | david@gmail.com         | 9000000007 |   42 | British     | 2026-03-20        |
|          8 | Emily Brown  | emily@gmail.com         | 9000000008 |   31 | American    | 2026-03-25        |
|          9 | Vikram Shetty | vikram@gmail.com        | 9000000009 |   26 | Indian      | 2026-04-01        |
|         10 | Priya Nair   | priya@gmail.com         | 9000000010 |   23 | Indian      | 2026-04-05        |
|         11 | Michael Chen | michael@gmail.com       | 9000000011 |   38 | Singaporean | 2026-04-10        |
|         12 | Kavya Menon  | kavya@gmail.com         | 9000000012 |   27 | Indian      | 2026-04-15        |
|         13 | Daniel Garcia | daniel@gmail.com        | 9000000013 |   34 | Spanish     | 2026-04-20        |
|         14 | Meera Iyer   | meera.visitor@gmail.com | 9000000014 |   29 | Indian      | 2026-04-25        |
+------------+--------------+-------------------------+------------+------+-------------+-------------------+
14 rows in set (0.00 sec)
```

The DELETE statement removes the selected visitor. The related visit rows are handled by the foreign-key ON DELETE CASCADE rule.

7.4 CORRELATED QUERY

```
mysql> SELECT s.staff_name, s.designation,
    ->        (SELECT COUNT(*)
    ->         FROM CONDITION_ASSESSMENT ca
    ->         WHERE ca.staff_id = s.staff_id) AS assessments,
    ->        (SELECT COUNT(*)
    ->         FROM RESTORATION_PROJECT p
    ->         WHERE p.manager_id = s.staff_id) AS projects_managed
    -> FROM STAFF s
    -> ORDER BY projects_managed DESC, assessments DESC;
+--------------+---------------------+-------------+------------------+
| staff_name   | designation         | assessments | projects_managed |
+--------------+---------------------+-------------+------------------+
| Rahul Sharma | Restoration Manager |           4 |                5 |
| Priya Menon  | Conservator         |           4 |                3 |
| Anita Rao    | Curator             |           2 |                0 |
| Meera Nair   | Archivist           |           1 |                0 |
| Sneha Iyer   | Curator             |           1 |                0 |
| Kiran Kumar  | Museum Director     |           0 |                0 |
| Vivek Rao    | Researcher          |           0 |                0 |
| Arjun Das    | Security Officer    |           0 |                0 |
| Divya Shah   | Archivist           |           0 |                0 |
| Manoj Kumar  | Technician          |           0 |                0 |
+--------------+---------------------+-------------+------------------+
10 rows in set (0.00 sec)
```

The inner queries refer to s.staff_id from the outer STAFF query. Therefore, the subqueries are correlated and are evaluated in relation to each staff row.

7.5 NESTED QUERY

```
mysql> SELECT artifact_id, artifact_name, estimated_value
    -> FROM ARTIFACT
    -> WHERE estimated_value >
    ->       (SELECT AVG(estimated_value) FROM ARTIFACT)
    -> ORDER BY estimated_value DESC;
+-------------+----------------------+-----------------+
| artifact_id | artifact_name        | estimated_value |
+-------------+----------------------+-----------------+
|          19 | Ancient Gold Coin    |       750000.00 |
|          17 | Ancient Bronze Vessel|       600000.00 |
|           7 | Royal Throne         |       550000.00 |
|          10 | Bharat Mata Painting |       500000.00 |
|           9 | Mythological Painting|       450000.00 |
|          13 | Temple Bronze Idol   |       400000.00 |
|          11 | Village Life Painting|       350000.00 |
|           2 | Granite Temple Statue|       320000.00 |
+-------------+----------------------+-----------------+
8 rows in set (0.00 sec)
```

The nested query calculates the average estimated value of all artifacts. The outer query returns artifacts whose estimated value is greater than that average.

Additional join query

```
mysql> SELECT a.artifact_id, a.artifact_name,
    ->        c.collection_name, m.museum_name
    -> FROM ARTIFACT a
    -> INNER JOIN COLLECTION c ON a.collection_id = c.collection_id
    -> INNER JOIN MUSEUM m ON c.museum_id = m.museum_id
    -> ORDER BY a.artifact_id;
+-------------+--------------------------+--------------------------+------------------------------+
| artifact_id | artifact_name            | collection_name          | museum_name                  |
+-------------+--------------------------+--------------------------+------------------------------+
|           1 | Ancient Stone Sculpture  | Ancient Indian Art       | National Heritage Museum     |
|           2 | Granite Temple Statue    | Ancient Indian Art       | National Heritage Museum     |
|           3 | Royal Manuscript         | Medieval Manuscripts     | National Heritage Museum     |
|           4 | Historical Document      | Medieval Manuscripts     | National Heritage Museum     |
|           5 | Royal Wooden Chest       | Royal Heritage           | Royal History Museum         |
|           6 | Royal Sword              | Royal Heritage           | Royal History Museum         |
|           7 | Royal Throne             | Mysuru Palace Collection | Royal History Museum         |
|           8 | Royal Jewellery Box      | Mysuru Palace Collection | Royal History Museum         |
|           9 | Mythological Painting    | Indian Paintings         | Indian Art Museum            |
|          10 | Bharat Mata Painting     | Indian Paintings         | Indian Art Museum            |
|          11 | Village Life Painting    | Indian Paintings         | Indian Art Museum            |
|          12 | Folk Art Composition     | Modern Art               | Indian Art Museum            |
|          13 | Temple Bronze Idol       | Temple Artifacts         | South Indian Heritage Museum |
|          14 | Temple Stone Panel       | Temple Artifacts         | South Indian Heritage Museum |
|          15 | Traditional Wooden Doll  | Traditional Crafts       | South Indian Heritage Museum |
|          16 | Handwoven Textile        | Traditional Crafts       | South Indian Heritage Museum |
|          17 | Ancient Bronze Vessel    | Archaeological Collection | Ancient Civilisation Museum |
|          18 | Ancient Terracotta Figure | Archaeological Collection | Ancient Civilisation Museum |
|          19 | Ancient Gold Coin        | Coins and Currency       | Ancient Civilisation Museum  |
|          20 | Silver Royal Coin        | Coins and Currency       | Ancient Civilisation Museum  |
+-------------+--------------------------+--------------------------+------------------------------+
20 rows in set (0.00 sec)
```

This query combines ARTIFACT, COLLECTION, and MUSEUM to show the collection and museum associated with each artifact.

8. STORED PROCEDURES, FUCNTIONS AND TRIGGERS

Stored routines encapsulate reusable database operations. Functions return a computed value, procedures execute a parameterized operation, and triggers execute automatically when a specified table event occurs.

8.1 STORED PROCEDURES OR FUNCTIONS

Stored Procedures:

Procedure: sp_artifacts_by_status

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE sp_artifacts_by_status(IN p_status VARCHAR(50))
    -> BEGIN
    ->     SELECT
    ->         artifact_id,
    ->         artifact_name,
    ->         artifact_type,
    ->         estimated_value,
    ->         status
    ->     FROM ARTIFACT
    ->     WHERE status = p_status
    ->     ORDER BY artifact_name;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL sp_artifacts_by_status('Needs Restoration');
+-------------+-----------------------+---------------+-----------------+--------------------+
| artifact_id | artifact_name         | artifact_type | estimated_value | status             |
+-------------+-----------------------+---------------+-----------------+--------------------+
|           2 | Granite Temple Statue | Sculpture     |       320000.00 | Needs Restoration  |
|          16 | Handwoven Textile     | Textile       |        95000.00 | Needs Restoration  |
|           9 | Mythological Painting | Painting      |       450000.00 | Needs Restoration  |
|           7 | Royal Throne          | Furniture     |       550000.00 | Needs Restoration  |
+-------------+-----------------------+---------------+-----------------+--------------------+
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.03 sec)
```

This procedure accepts a status value and returns artifacts matching that status, sorted by artifact name.

Procedure: sp_artifact_restoration_projects

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE sp_artifact_restoration_projects(IN p_artifact_id INT)
    -> BEGIN
    ->     SELECT
    ->         project_id,
    ->         project_name,
    ->         objective,
    ->         start_date,
    ->         end_date,
    ->         status,
    ->         cost
    ->     FROM RESTORATION_PROJECT
    ->     WHERE artifact_id = p_artifact_id
    ->     ORDER BY start_date;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL sp_artifact_restoration_projects(4);
+------------+-----------------------------+-----------------------------------------------------+------------+------------+-----------+----------+
| project_id | project_name                | objective                                           | start_date | end_date   | status    | cost     |
+------------+-----------------------------+-----------------------------------------------------+------------+------------+-----------+----------+
|          8 | Historical Document Preservation | Preserve and digitally archive the historical document. | 2026-01-10 | 2026-04-30 | Completed | 30000.00 |
+------------+-----------------------------+-----------------------------------------------------+------------+------------+-----------+----------+
1 row in set (0.00 sec)
```

This procedure returns restoration projects associated with a specified artifact in start-date order.

Functions:

Function: fn_artifact_age

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION fn_artifact_age(p_artifact_id INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE v_age INT;
    ->
    ->     SELECT TIMESTAMPDIFF(
    ->         YEAR,
    ->         creation_date,
    ->         CURDATE()
    ->     )
    ->     INTO v_age
    ->     FROM ARTIFACT
    ->     WHERE artifact_id = p_artifact_id;
    ->
    ->     RETURN v_age;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> SELECT
    ->     artifact_id,
    ->     artifact_name,
    ->     fn_artifact_age(artifact_id) AS Artifact_Age
    -> FROM ARTIFACT;
+-------------+--------------------------+--------------+
| artifact_id | artifact_name            | Artifact_Age |
+-------------+--------------------------+--------------+
|           1 | Ancient Stone Sculpture  |          776 |
|           2 | Granite Temple Statue    |          726 |
|           3 | Royal Manuscript         |          376 |
|           4 | Historical Document      |          326 |
|           5 | Royal Wooden Chest       |          246 |
|           6 | Royal Sword              |          251 |
|           7 | Royal Throne             |          226 |
|           8 | Royal Jewellery Box      |          206 |
|           9 | Mythological Painting    |          136 |
|          10 | Bharat Mata Painting     |          121 |
|          11 | Village Life Painting    |          106 |
|          12 | Folk Art Composition     |           91 |
|          13 | Temple Bronze Idol       |          576 |
|          14 | Temple Stone Panel       |          526 |
|          15 | Traditional Wooden Doll  |          146 |
|          16 | Handwoven Textile        |          126 |
|          17 | Ancient Bronze Vessel    |          926 |
|          18 | Ancient Terracotta Figure |         1026 |
|          19 | Ancient Gold Coin        |          376 |
|          20 | Silver Royal Coin        |          326 |
+-------------+--------------------------+--------------+
20 rows in set (0.00 sec)
```

This function calculates the age of an artifact in completed years using its creation_date and the current date.

Function: fn_total_restoration_cost

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION fn_total_restoration_cost(p_artifact_id INT)
    -> RETURNS DECIMAL(12,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE v_total DECIMAL(12,2);
    ->
    ->     SELECT COALESCE(SUM(cost), 0.00)
    ->     INTO v_total
    ->     FROM RESTORATION_PROJECT
    ->     WHERE artifact_id = p_artifact_id;
    ->
    ->     RETURN v_total;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

This function returns the sum of restoration project costs for a given artifact. COALESCE returns 0.00 when no project cost is found.

8.2 TRIGGERS

Trigger: material cost calculation

```
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER trg_material_before_insert
    -> BEFORE INSERT ON RESTORATION_MATERIAL
    -> FOR EACH ROW
    -> BEGIN
    ->     SET NEW.total_cost = NEW.unit_cost * NEW.quantity_used;
    -> END$$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> INSERT INTO RESTORATION_MATERIAL
    -> (material_id, material_name, material_type, supplier,
    ->  unit, unit_cost, quantity_used, total_cost)
    -> VALUES
    -> (11, 'Conservation Wax', 'Protective Material',
    ->  'Heritage Supplies', 'kg', 1500.00, 4.00, 0);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT material_id, material_name, unit_cost,
    ->        quantity_used, total_cost
    -> FROM RESTORATION_MATERIAL
    -> WHERE material_id = 11;
+-------------+------------------+-----------+---------------+------------+
| material_id | material_name    | unit_cost | quantity_used | total_cost |
+-------------+------------------+-----------+---------------+------------+
|          11 | Conservation Wax |   1500.00 |          4.00 |    6000.00 |
+-------------+------------------+-----------+---------------+------------+
1 row in set (0.00 sec)
```

The trigger calculates total_cost automatically before a new restoration material row is inserted. The runnable application schema also contains a BEFORE UPDATE trigger so the calculated value is refreshed when unit_cost or quantity_used changes.

Trigger: exhibition date validation

```
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER trg_exhibition_before_insert
    -> BEFORE INSERT ON EXHIBITION
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.end_date < NEW.start_date THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Exhibition end date cannot be before start date';
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> DELIMITER ;
mysql> INSERT INTO EXHIBITION
    -> (exhibition_id, museum_id, exhibition_name,
    ->  description, theme, start_date, end_date, status)
    -> VALUES
    -> (9, 1, 'Invalid Exhibition',
    ->  'Testing date validation',
    ->  'Test',
    ->  '2026-12-01',
    ->  '2026-10-01',
    ->  'Planned');
ERROR 1644 (45000): Exhibition end date cannot be before start date
```

The trigger rejects an exhibition whose end date is earlier than its start date. The runnable application schema also contains a corresponding BEFORE UPDATE trigger.

The DB Features interface provides trigger demonstrations. The backend runs the demonstration in a transaction and rolls it back, so test rows are not retained.

9. FRONT END DEVELOPEMNT

9.1 Frontend

The frontend of the Digital Museum Artifact & Restoration Management System is developed using Next.js 15, React 19 and TypeScript. It provides an interactive, user-friendly interface through which administrators and staff can access the system's features.

The frontend allows users to log in, navigate the dashboard, manage museum-related information, interact with database operations and access the MongoDB Studio section. It also provides separate data-entry forms for the five MongoDB collections:

- Artifact Research
- Conservation Observations
- Historical References
- Exhibition Feedback
- Digital Documentation

The frontend includes a stored-procedure interface where users can select an artifact status or enter an artifact ID and execute the corresponding procedure. The results are displayed on the page, along with appropriate messages when no records are found or an error occurs.

Technologies used:

- Next.js 15
- React 19
- TypeScript
- HTML and CSS

9.2 Backend

The backend is implemented using Next.js API Route Handlers, which process requests from the frontend and communicate with the databases. It handles authentication, database operations, user management and the execution of SQL routines.

The application uses MySQL as its relational database to store structured information about artifacts, collections, staff, exhibitions, visitors and restoration projects. The MySQL connection is managed using the mysql2 package.

The backend also integrates MongoDB using the MongoDB Node.js driver. It manages flexible document-based information through five separate collections, allowing users to save and retrieve research, conservation observations, historical references, exhibition feedback and digital documentation.

Authentication is handled using bcryptjs for password hashing and HTTP-only session cookies to maintain user sessions. The backend also supports the execution of database triggers, stored functions and stored procedures.

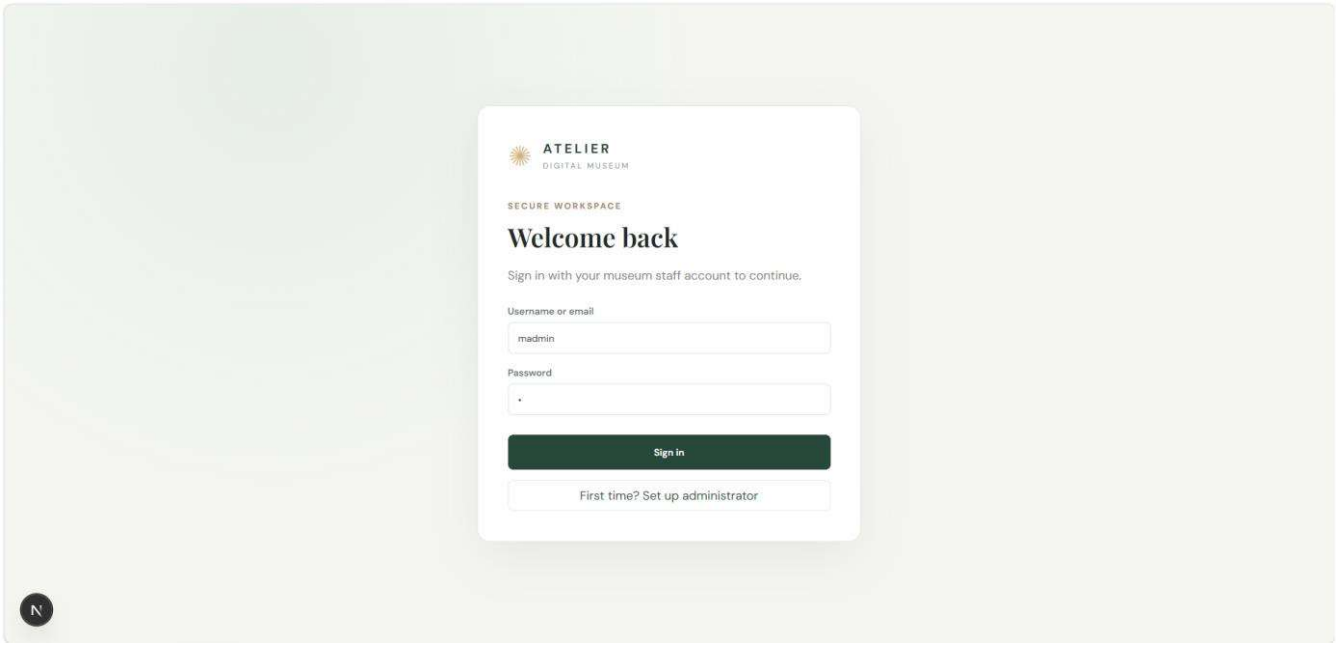
Technologies used:

- Next.js API Route Handlers
- Node.js
- TypeScript
- MySQL with mysql2
- MongoDB with the MongoDB Node.js driver
- bcryptjs for password hashing

9.3 Frontend–Backend Communication

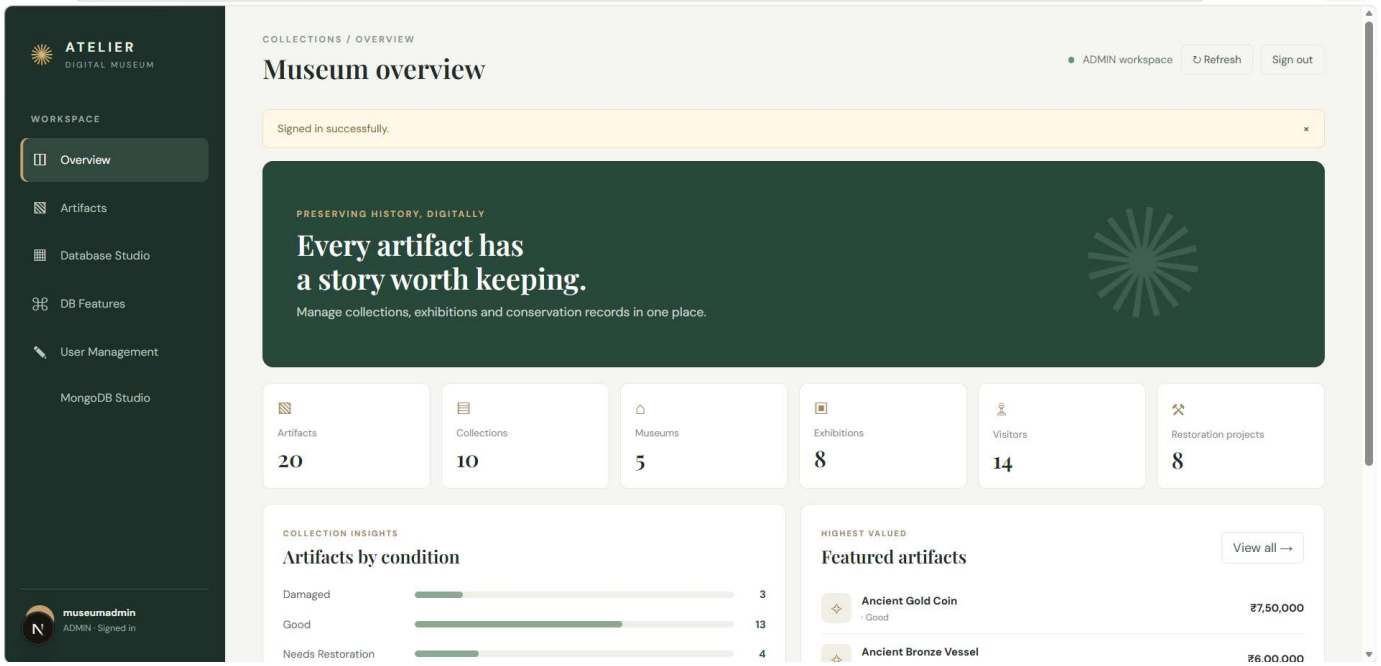
The frontend and backend communicate through API requests. When a user performs an operation, such as saving a document or running a stored procedure, the frontend sends a request to the appropriate API endpoint. The backend processes the request, interacts with MySQL or MongoDB, and returns the result in JSON format. The frontend then displays the response to the user.

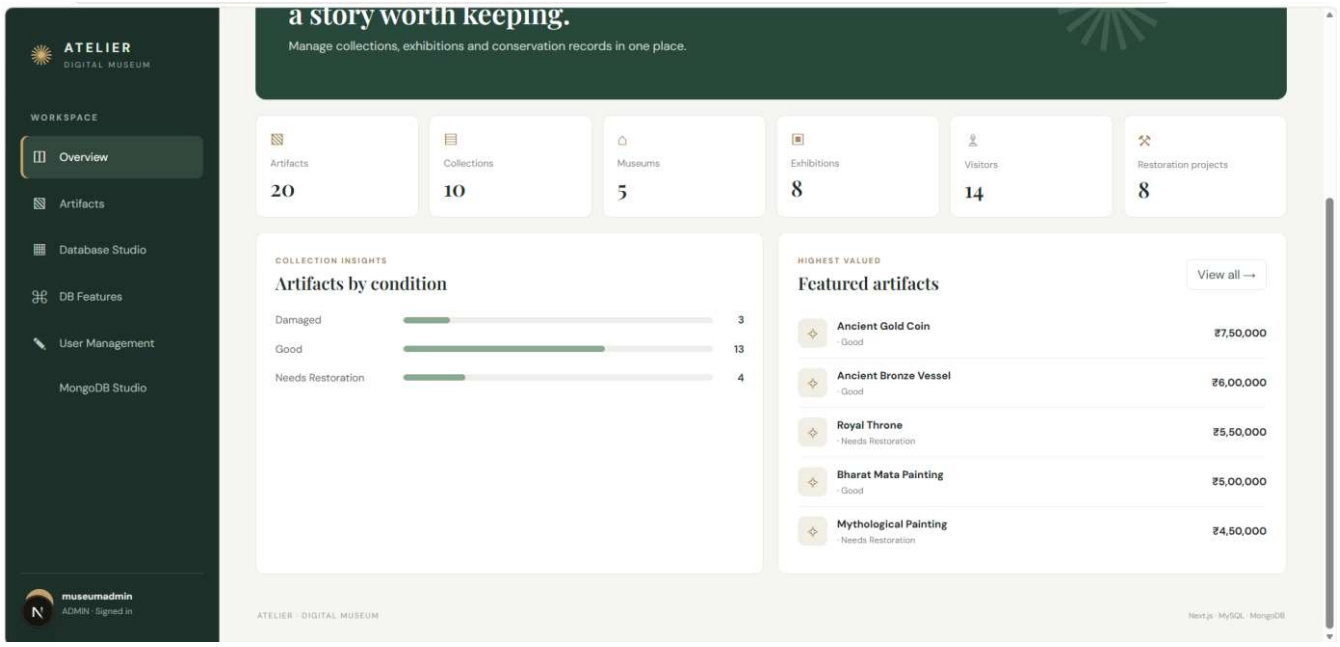
Login:



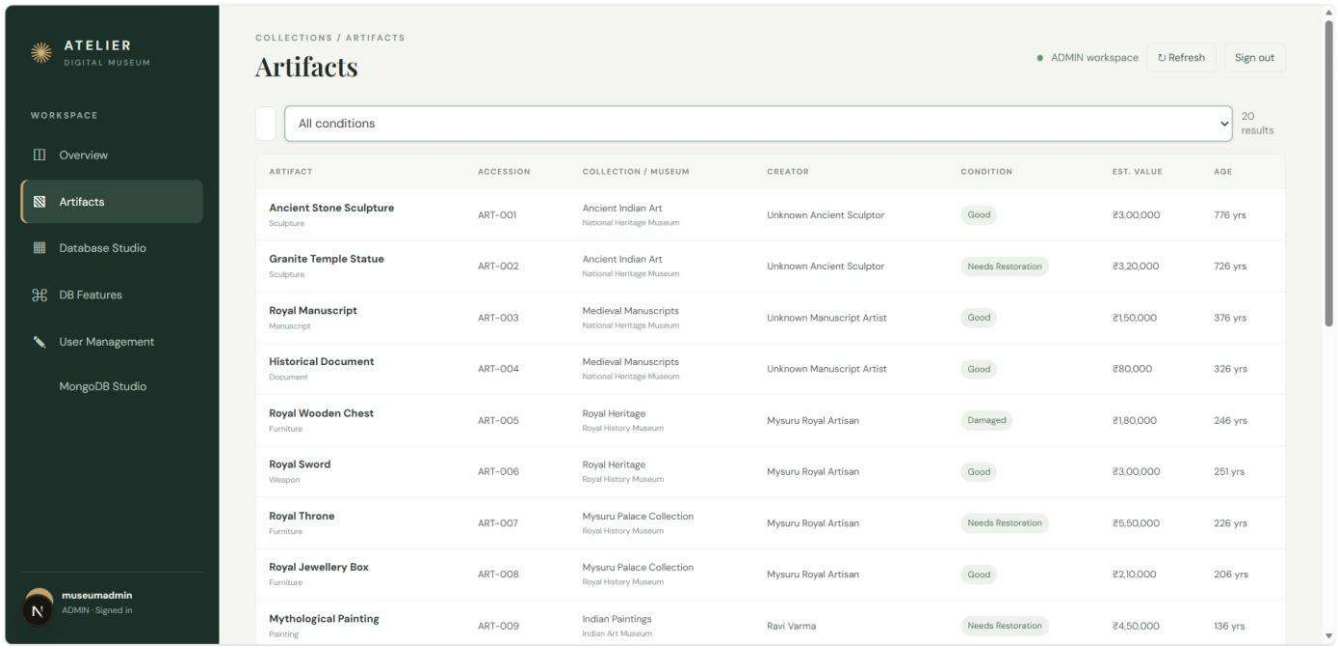
Interface module	Implemented functionality
Overview	Displays a dashboard for accessing the main museum management features and viewing available system information.
Artifacts	Provides access to artifact information and related museum records.
Database Studio	Provides database operations for managing MySQL records, including adding, editing, deleting, searching and viewing data.
DB Features	Demonstrates MySQL database features, including triggers, stored functions and stored procedures. Users can execute procedures to retrieve artifacts by status and restoration projects by artifact ID.
User Management	Allows administrators to manage user accounts and staff access to the application.
MongoDB Studio	Provides separate data-entry forms and document management for five MongoDB collections: artifact research, conservation observations, historical references, exhibition feedback and digital documentation.

Dashboard:

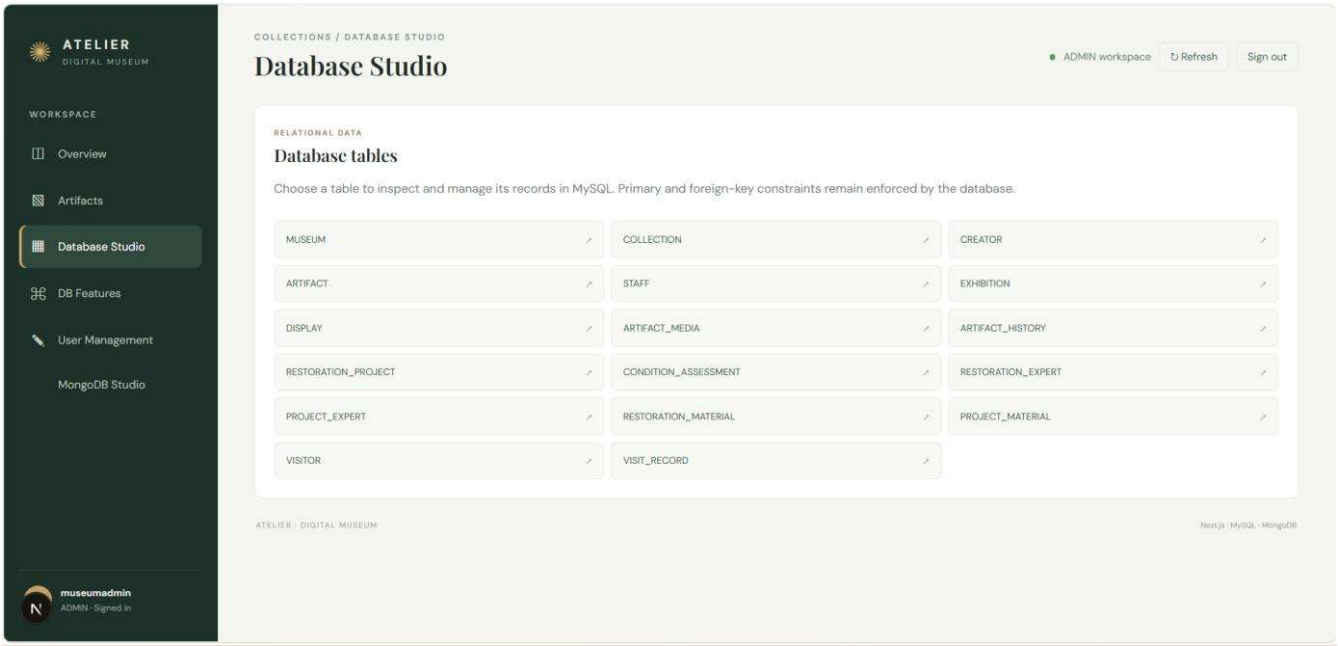

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Overview
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MongoDB Studio
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ADMIN · Signed in
COLLECTIONS / OVERVIEW
Museum overview
ADMIN workspace
Refresh
Sign out
Signed in successfully.
PRESERVING HISTORY, DIGITALLY
Every artifact has
a story worth keeping.
Manage collections, exhibitions and conservation records in one place.
Artifacts 20
Collections 10
Museums 5
Exhibitions 8
Visitors 14
Restoration projects 8
COLLECTION INSIGHTS
Artifacts by condition
Damaged 3
Good 13
Needs Restoration 4
HIGHEST VALUED
Featured artifacts
View all →
Ancient Gold Coin · Good ₹7,50,000
Ancient Bronze Vessel


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a story worth keeping.
Manage collections, exhibitions and conservation records in one place.
Artifacts 20
Collections 10
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COLLECTION INSIGHTS
Artifacts by condition
Damaged 3
Good 13
Needs Restoration 4
HIGHEST VALUED
Featured artifacts
View all →
Ancient Gold Coin · Good ₹7,50,000
Ancient Bronze Vessel · Good ₹6,00,000
Royal Throne · Needs Restoration ₹5,50,000
Bharat Mata Painting · Good ₹5,00,000
Mythological Painting · Needs Restoration ₹4,50,000
ATELIER · DIGITAL MUSEUM
Next.js · MySQL · MongoDB

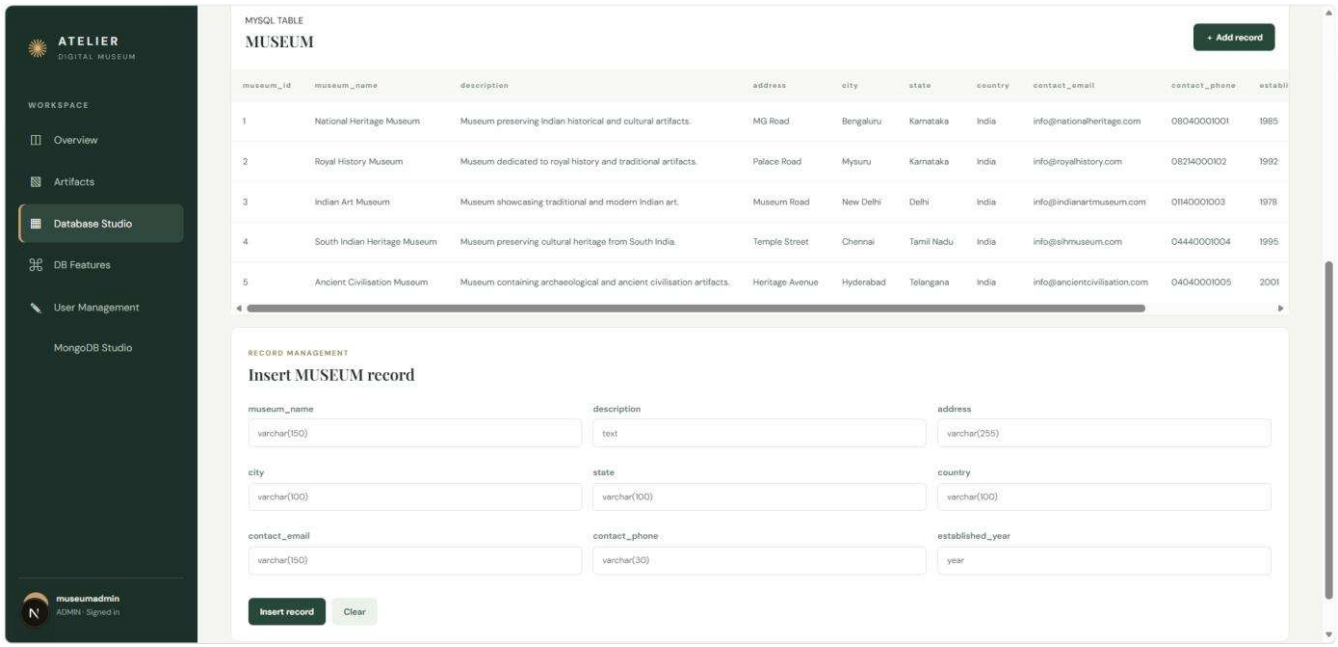
Artifacts:



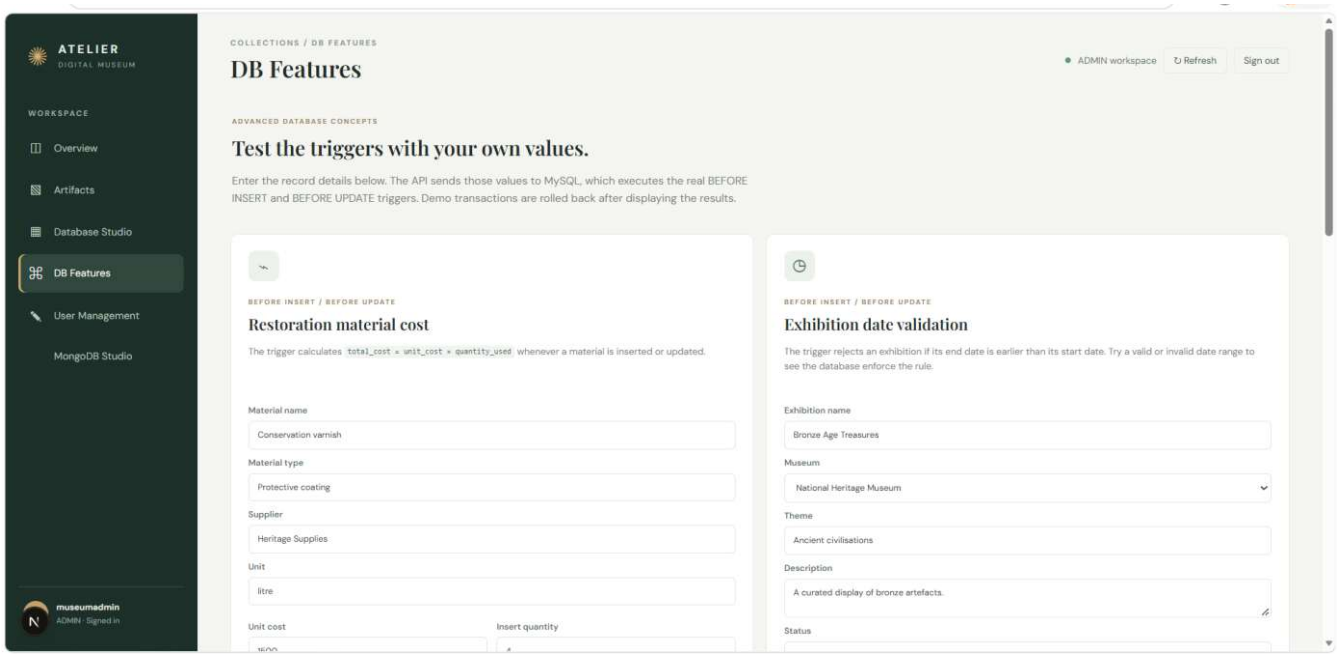
Database Studio:



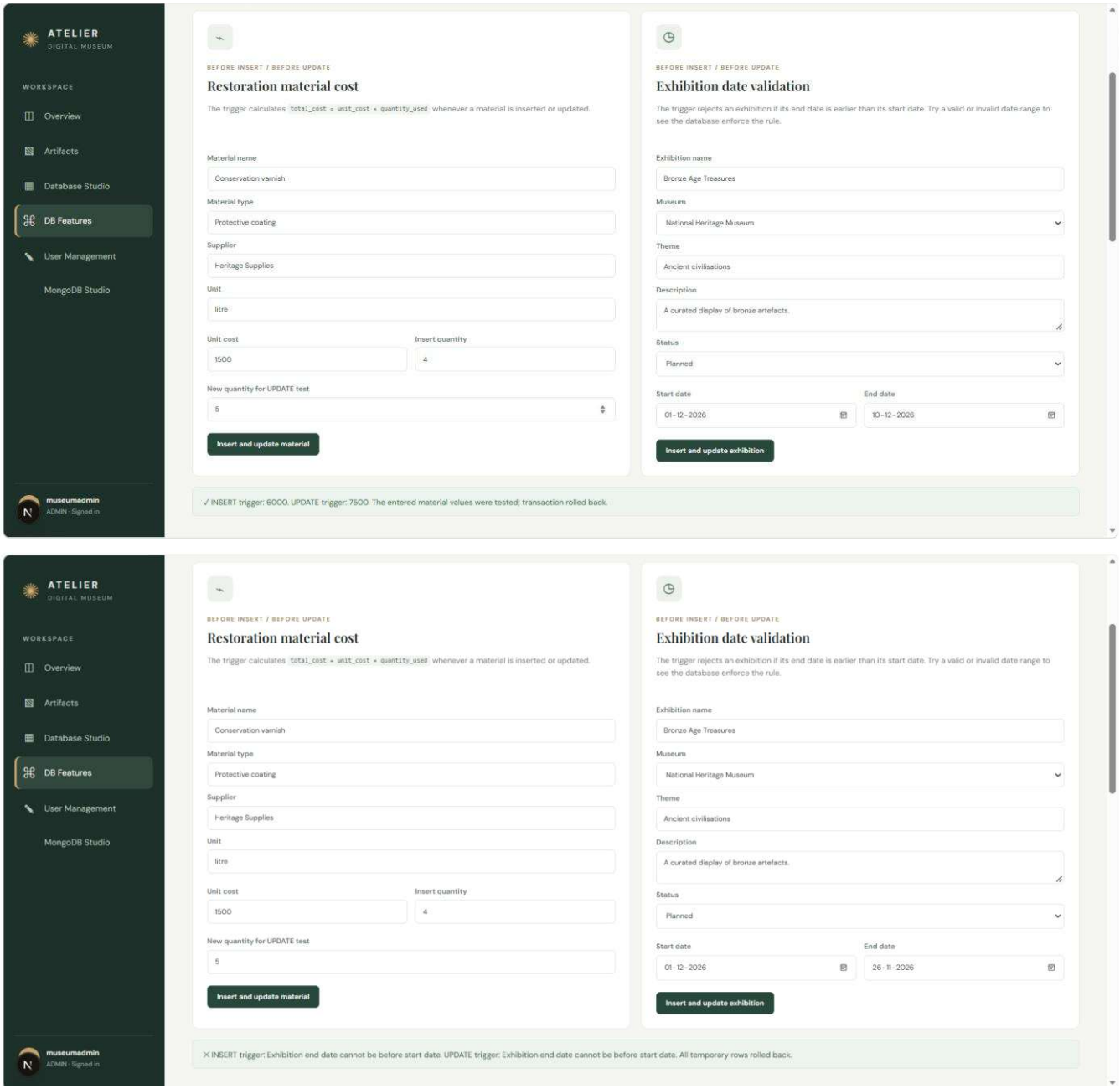
We can view each table and add a new record



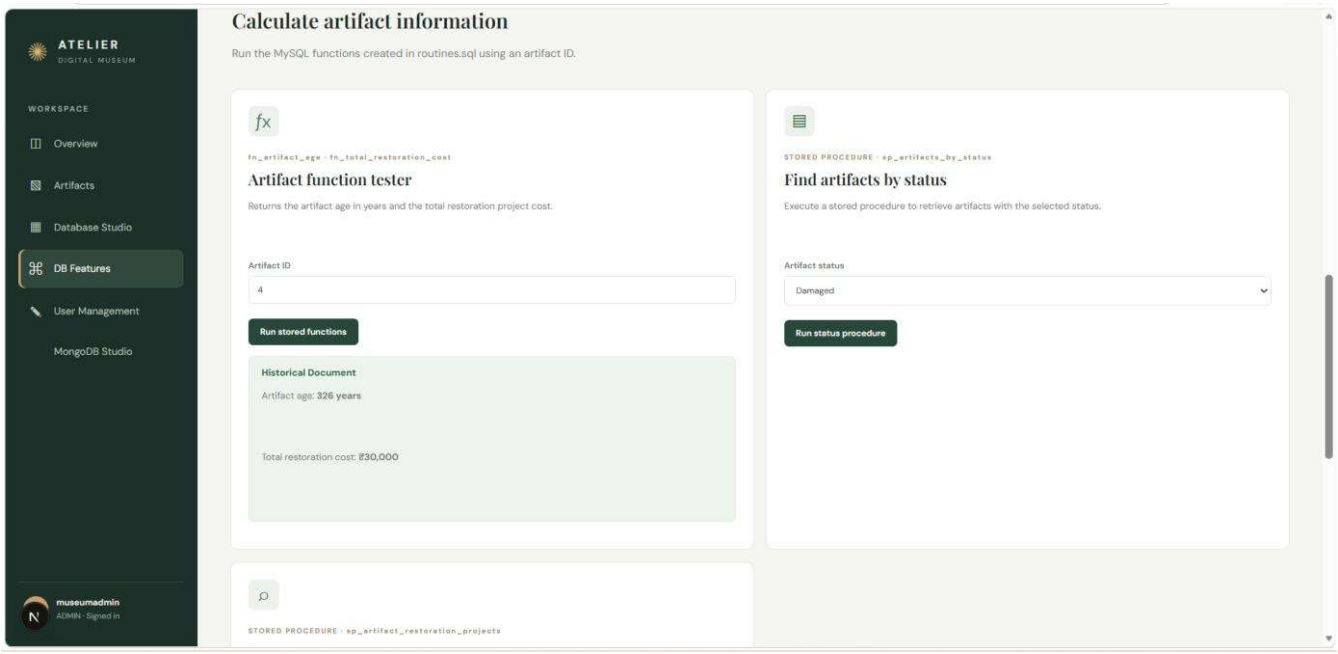
DB features:



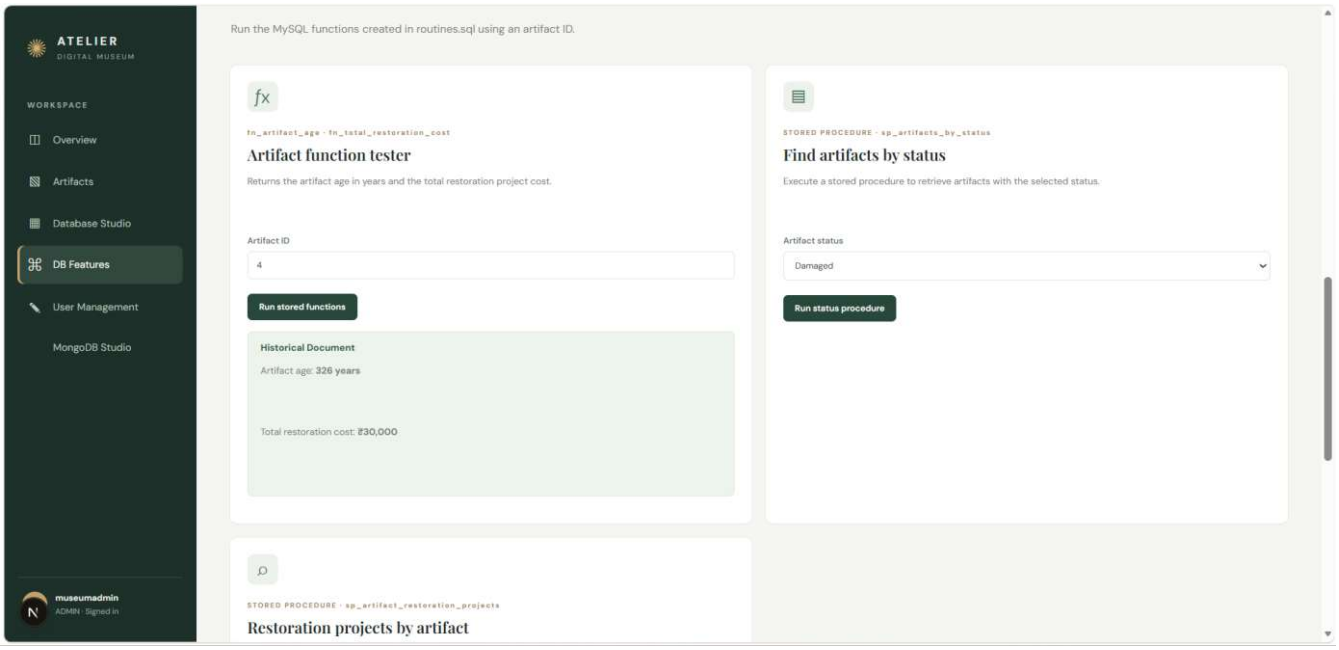
Testing the triggers

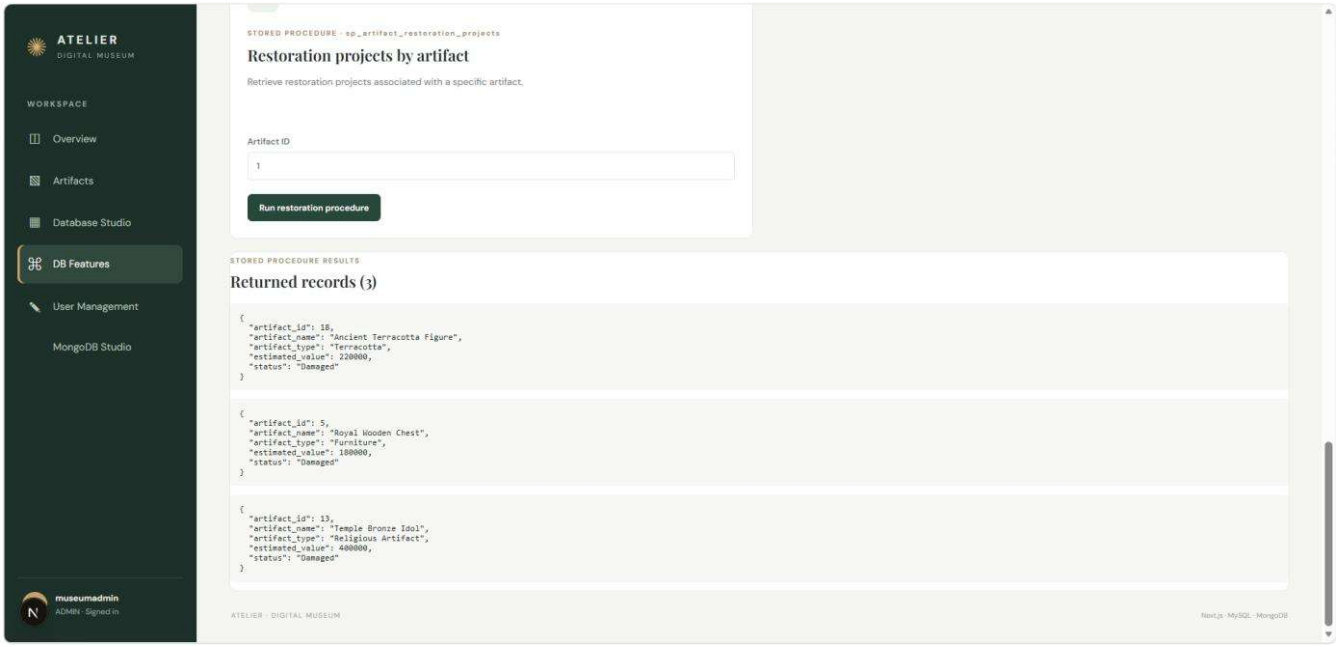


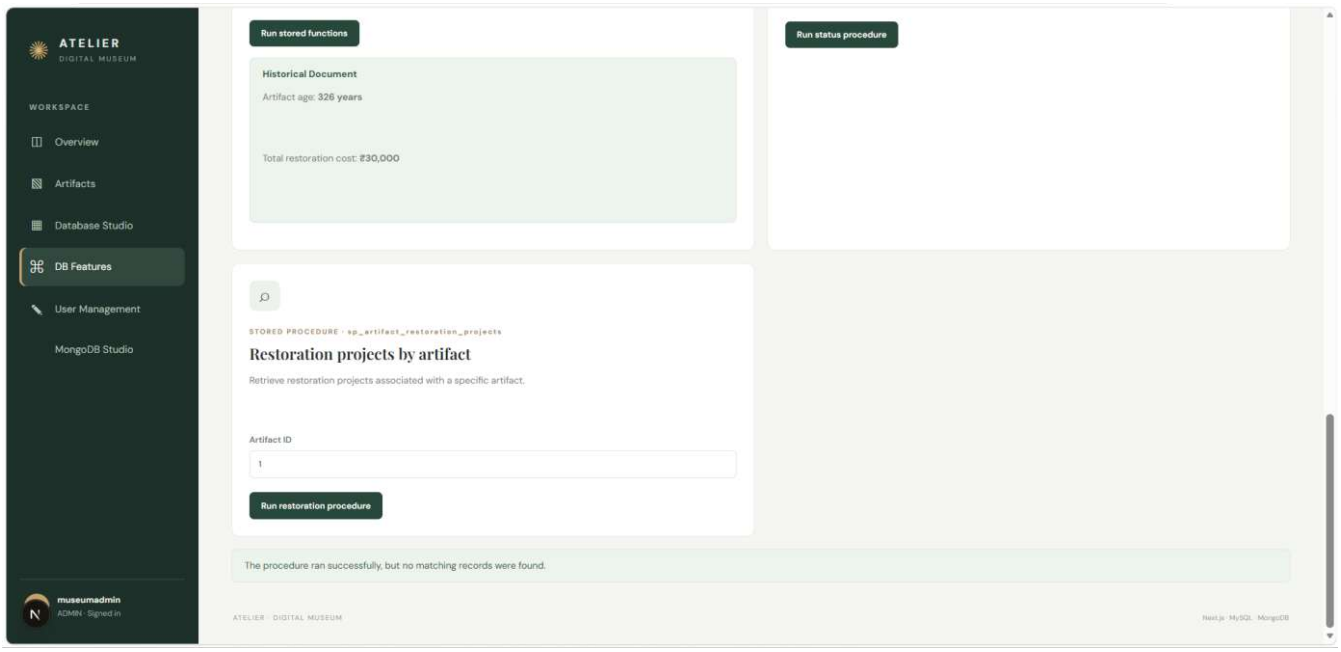
Testing the functions



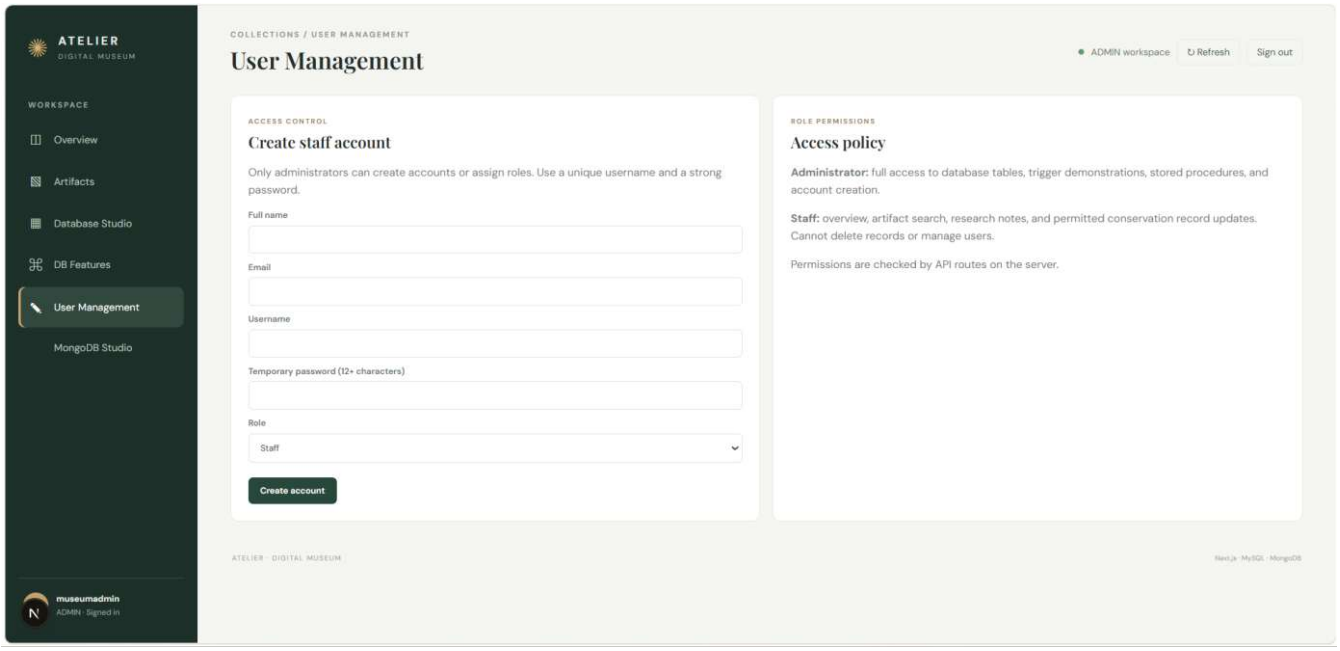
Testing the procedures


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Run the MySQL functions created in routines.sql using an artifact ID.
fn_artifact_age · fn_total_restoration_cost
Artifact function tester
Returns the artifact age in years and the total restoration project cost.
Artifact ID
4
Run stored functions
Historical Document
Artifact age: 326 years
Total restoration cost: ₹30,000
STORED PROCEDURE · sp_artifacts_by_status
Find artifacts by status
Execute a stored procedure to retrieve artifacts with the selected status.
Artifact status
Damaged
Run status procedure
STORED PROCEDURE · sp_artifact_restoration_projects
Restoration projects by artifact

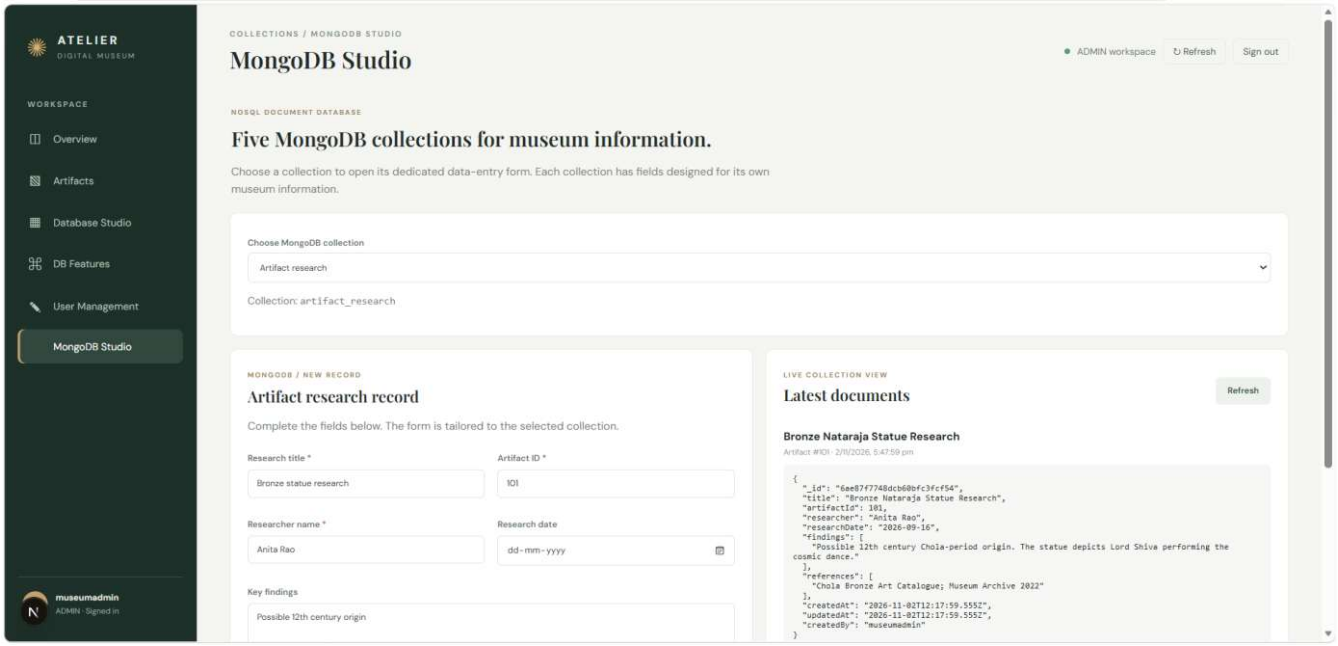

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STORED PROCEDURE · sp_artifact_restoration_projects
Restoration projects by artifact
Retrieve restoration projects associated with a specific artifact.
Artifact ID
1
Run restoration procedure
STORED PROCEDURE RESULTS
Returned records (3)
{
"artifact_id": 18,
"artifact_name": "Ancient Terracotta Figure",
"artifact_type": "Terracotta",
"estimated_value": 220000,
"status": "Damaged"
}
{
"artifact_id": 5,
"artifact_name": "Royal Wooden Chest",
"artifact_type": "Furniture",
"estimated_value": 180000,
"status": "Damaged"
}
{
"artifact_id": 13,
"artifact_name": "Temple Bronze Idol",
"artifact_type": "Religious Artifact",
"estimated_value": 400000,
"status": "Damaged"
}
ATELIER · DIGITAL MUSEUM
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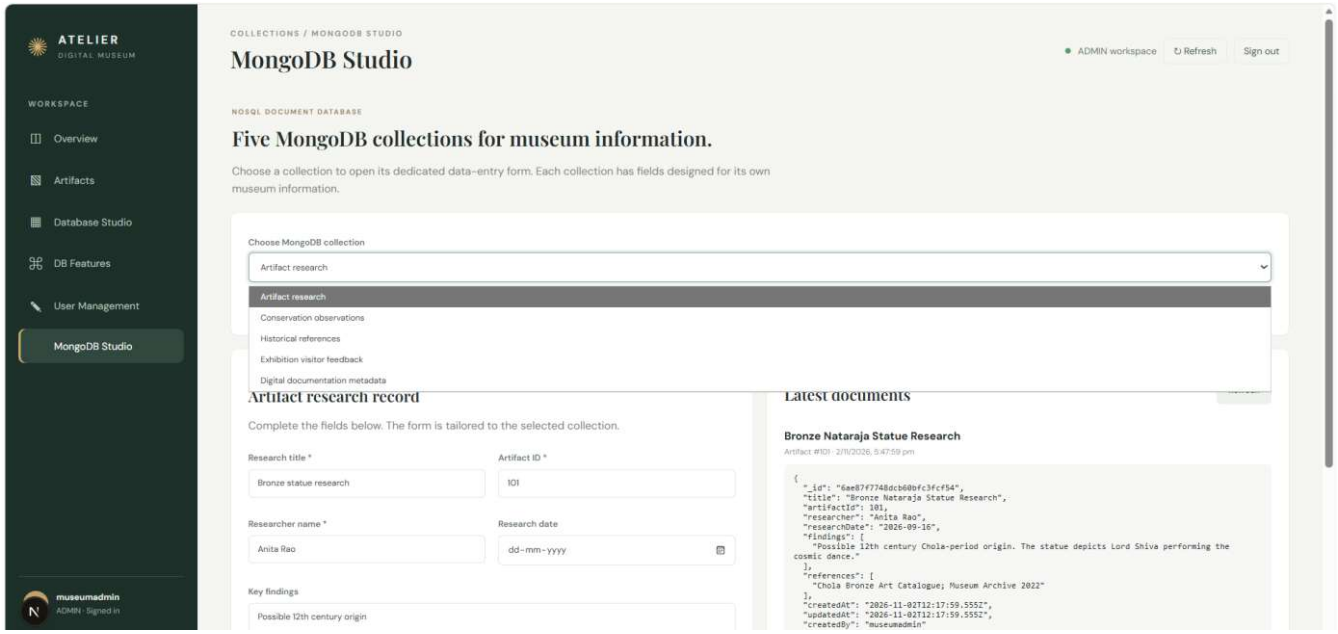


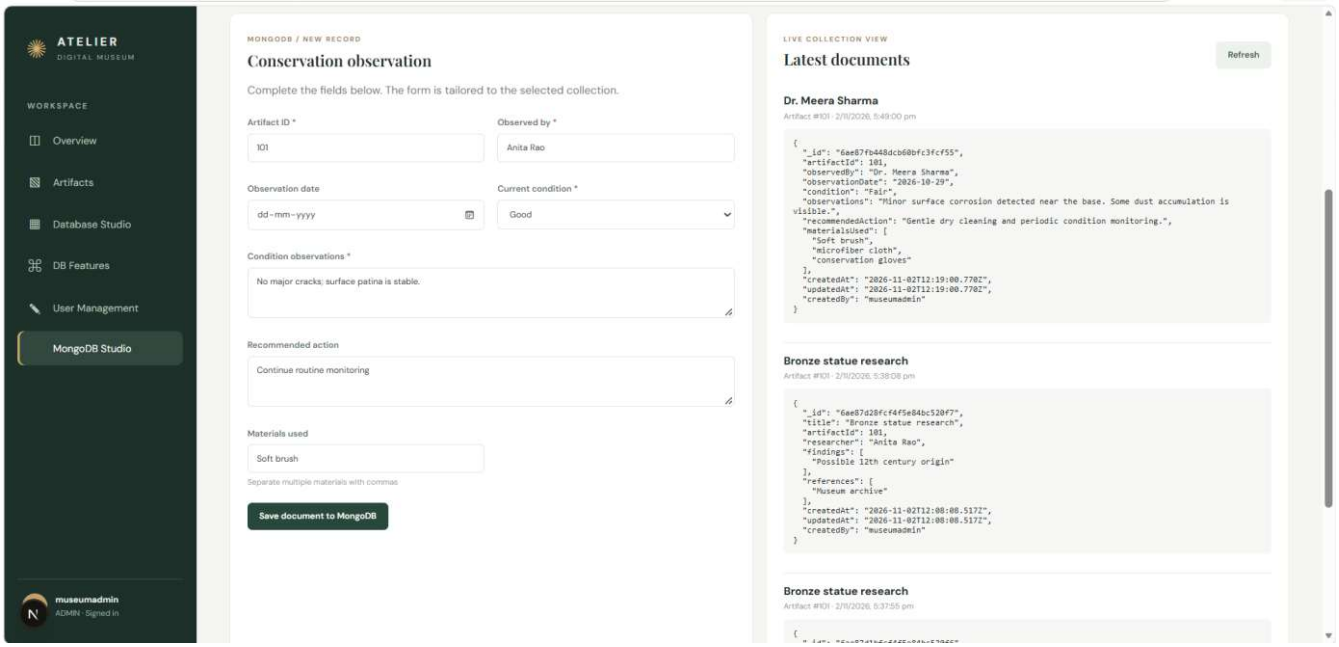
User Management:



MongoDB Studio:


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Sign out
NOSQL DOCUMENT DATABASE
Five MongoDB collections for museum information.
Choose a collection to open its dedicated data-entry form. Each collection has fields designed for its own museum information.
Choose MongoDB collection
Artifact research
Collection: artifact_research
MONGODB / NEW RECORD
Artifact research record
Complete the fields below. The form is tailored to the selected collection.
Research title *
Bronze statue research
Artifact ID *
101
Researcher name *
Anita Rao
Research date
dd-mm-yyyy
Key findings
Possible 12th century origin
LIVE COLLECTION VIEW
Latest documents
Refresh
Bronze Nataraja Statue Research
Artifact #101 · 2/11/2026, 5:47:59 pm
{
"_id": "6ae87f7748dcb60bfc3fcf54",
"title": "Bronze Nataraja Statue Research",
"artifactId": 101,
"researcher": "Anita Rao",
"researchDate": "2026-09-16",
"findings": [
"Possible 12th century Chola-period origin. The statue depicts Lord Shiva performing the cosmic dance."
],
"references": [
"Chola Bronze Art Catalogue; Museum Archive 2022"
],
"createdAt": "2026-11-02T12:17:59.555Z",
"updatedAt": "2026-11-02T12:17:59.555Z",
"createdBy": "museumadmin"
}


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NOSQL DOCUMENT DATABASE
Five MongoDB collections for museum information.
Choose a collection to open its dedicated data-entry form. Each collection has fields designed for its own museum information.
Choose MongoDB collection
Artifact research
Artifact research
Conservation observations
Historical references
Exhibition visitor feedback
Digital documentation metadata
Artifact research record
Complete the fields below. The form is tailored to the selected collection.
Research title *
Bronze statue research
Artifact ID *
101
Researcher name *
Anita Rao
Research date
dd-mm-yyyy
Key findings
Possible 12th century origin
Latest documents
Bronze Nataraja Statue Research
Artifact #101 · 2/11/2026, 5:47:59 pm
{
"_id": "6ae87f7748dcb60bfc3fcf54",
"title": "Bronze Nataraja Statue Research",
"artifactId": 101,
"researcher": "Anita Rao",
"researchDate": "2026-09-16",
"findings": [
"Possible 12th century Chola-period origin. The statue depicts Lord Shiva performing the cosmic dance."
],
"references": [
"Chola Bronze Art Catalogue; Museum Archive 2022"
],
"createdAt": "2026-11-02T12:17:59.555Z",
"updatedAt": "2026-11-02T12:17:59.555Z",
"createdBy": "museumadmin"



9.4 MONGODB DOCUMENT COLLECTIONS

MongoDB complements the relational database by storing flexible document records. The application includes a MongoDB Studio area with a separate collection selection and collection-specific data-entry form for each of the following collections:

artifact_research — research findings, notes, tags and references for artifacts.

conservation observations — condition and conservation observations, including observer, condition, observations and severity.

historical_references — archival or bibliographic sources, authors, publication years and URLs.

exhibition_feedback — exhibition ratings and visitor feedback.

digital_documentation — digital asset metadata such as file name, MIME type, storage path and image dimensions.

The application accepts valid JSON documents. It adds createdAt, updatedAt and createdBy metadata when saving. The five collections are created automatically when their first document is saved. The MongoDB feature stores metadata and documents; the final package does not implement vector embeddings or vector search. Existing MySQL tables and records are not modified by saving MongoDB documents.

9.5 WEB APPLICATION AND USER INTERFACE

The application is a Next.js web application. The main page is app/page.tsx, the shared stylesheet is app/globals.css, and the root layout is app/layout.tsx. API functionality is implemented in app/api route handlers. Database access and authentication helpers are in the lib folder.

Authentication supports Admin and Staff roles. Password verification uses bcryptjs, and session cookies are HTTP-only. Protected API operations check the authenticated user's role where required. The user-management API is used for staff account administration.

The application includes a dashboard/workspace for museum operations, MySQL-backed data and routine features, and MongoDB Studio for the five document collections. The routines interface is intended to demonstrate the two stored procedures. The corrected status-procedure UI reads available statuses from the database and provides feedback for results, empty results and errors.

9.6 TESTING AND VALIDATION

Database testing should verify that the configured MySQL database is reachable and that the expected tables and routines exist. In MySQL Workbench, select the configured database before running inspection commands. The procedures can be tested directly with CALL sp artifacts by status('Good') and CALL sp_artifact_restoration_projects(1), substituting a status and artifact identifier that exist in the current data.

Application testing should include signing in, checking role-protected areas, loading the routine interface, running the status procedure with a status present in ARTIFACT, running the restoration-project procedure with a valid artifact identifier, and checking the displayed result or empty-result message. Also test saving a valid JSON document into each MongoDB collection and confirm that the saved document appears in the collection listing.

The project archive was inspected for its source files and SQL routine definitions. Successful execution against a user's local MySQL and MongoDB servers depends on the user's environment, credentials, database contents and installed services; this report does not claim that those live services were independently tested.

9.7 Artifact details and database routines

The artifact overview endpoint calls fn_artifact_age() and fn_total_restoration_cost() to enrich artifact rows. The artifact detail endpoint retrieves related project, expert, material, condition assessment, history, display, and media information. The application also exposes report endpoints for the predefined SQL reports and demonstration endpoints for the triggers.

9.4 Running the application

1. Extract the final ZIP archive and open the inner digital_museum_next folder in Visual Studio Code. This is the folder containing package.json.

2. Create a local .env.local file using .env.example as a guide. Configure MYSQL_HOST, MYSQL_PORT, MYSQL_USER, MYSQL_PASSWORD and MYSQL_DATABASE. For the existing database, use MYSQL_DATABASE=digital_museum_dbs if that is the database name in MySQL Workbench.

3. Configure MONGODB_URI and MONGODB_DATABASE. The local development values can be mongodb://127.0.0.1:27017 and digital_museum when MongoDB is running locally.

4. Set SESSION_SECRET and SETUP_SECRET to long, private values. Do not publish .env.local or commit it to a public repository.

5. Open a terminal in the digital_museum_next folder. If PowerShell blocks npm.ps1, run Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass in that terminal session, then continue.

6. Run npm install, followed by npm run dev. Keep the terminal open. Open the local URL printed by Next.js; if port 3000 is occupied, Next.js may use another port such as 3001.

7. Ensure MySQL and MongoDB are running and that the environment variables match the local database configuration. Sign in with a configured account. Use the Admin account to manage staff accounts where the user-management feature is available.

REFERENCES

1. MySQL 8.0 Reference Manual, https://dev.mysql.com/doc/refman/8.0/en/
2. Next.js Documentation, https://nextjs.org/docs
3. React Documentation, https://react.dev/learn
4. MongoDB Node.js Driver Documentation, https://www.mongodb.com/docs/drivers/node/current/
5. TypeScript Documentation, https://www.typescriptlang.org/docs/
6. bcryptjs package documentation, https://www.npmjs.com/package/bcryptjs
7. Final project source archive: Digital Museum Final.zip (app/, lib/, database/, package.json and README.md).

APPENDIX A: DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Term	Definition
DBMS	Database Management System.
DDL	Data Definition Language; commands used to define database structures.
DML	Data Manipulation Language; commands used to insert, update, and delete records.
ER	Entity–Relationship model used to represent entities and their relationships.
PK	Primary Key; uniquely identifies a row in a relation.
FK	Foreign Key; references a key in another relation.
UQ	Unique constraint; prevents duplicate values in a column or column set.
CRUD	Create, Read, Update, Delete operations.
API	Application Programming Interface.
SQL	Structured Query Language.
Flask	Python web framework used for the backend.
PyMySQL	Python library used to connect to MySQL.
Trigger	A database program automatically executed on a table event.
Stored procedure	A named SQL routine invoked using CALL.
Stored function	A named SQL routine that returns a value.

GitHub Link:

https://github.com/riona-ds06/Digital-Museum-Artifact-Restoration-Management-System-.git